

N706

Hardware User Guide

Issue 1.2 Date 2022-08-02



Copyright © Neoway Technology Co., Ltd 2022. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Neoway Technology Co., Ltd.

neoway is the trademark of Neoway Technology Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

This document provides guide for users to use N706.

This document is intended for system engineers (SEs), development engineers, and test engineers.

THIS GUIDE PROVIDES INSTRUCTIONS FOR CUSTOMERS TO DESIGN THEIR APPLICATIONS. PLEASE FOLLOW THE RULES AND PARAMETERS IN THIS GUIDE TO DESIGN AND COMMISSION. NEOWAY WILL NOT TAKE ANY RESPONSIBILITY OF BODILY HURT OR ASSET LOSS CAUSED BY IMPROPER OPERATIONS.

THE INFORMATION IN THIS DOCUMENT IS SUBJECT TO CHANGE WITHOUT NOTICE DUE TO PRODUCT VERSION UPDATE OR OTHER REASONS.

EVERY EFFORT HAS BEEN MADE IN PREPARATION OF THIS DOCUMENT TO ENSURE ACCURACY OF THE CONTENTS, BUT ALL STATEMENTS, INFORMATION, AND RECOMMENDATIONS IN THIS DOCUMENT DO NOT CONSTITUTE A WARRANTY OF ANY KIND, EXPRESS OR IMPLIED.

Neoway provides customers complete technical support. If you have any question, please contact your account manager or email to the following email addresses:

Sales@neoway.com

Support@neoway.com

Website: <http://www.neoway.com>

Contents

About This Document.....	viii
1 Safety Recommendations	11
2 About N706	12
2.1 Product Overview.....	12
2.2 Block Diagram.....	13
2.3 Basic Features	13
3 Reference Standards	15
4 Module Pins	16
4.1 Pin Layout	16
4.2 Pin Descriptions	17
5 Application Interfaces	24
5.1 Power Interfaces	24
5.1.1 VBAT.....	24
5.1.2 VDD_1P8.....	28
5.1.3 VRTC.....	29
5.2 Control Interfaces.....	29
5.2.1 Startup	30
5.2.2 Shutdown.....	32
5.2.3 Reset	33
5.2.4 SLEEP/WAKEUP	35
5.3 Peripheral Interfaces.....	38
5.3.1 USB	39
5.3.2 UART	40
5.3.3 USIM.....	44
5.3.4 SPI.....	46
5.3.5 I2C	47
5.4 Audio Interface	48
5.5 Video Interfaces	49
5.5.1 LCD	49
5.5.2 Camera.....	50
5.6 Other Interfaces	50
5.6.1 ADC	50
5.6.2 USB_BOOT	51
5.6.3 NET_LIGHT	51
5.6.4 RING.....	52
5.6.5 GPIO.....	53
5.7 RF Interfaces.....	53
5.7.1 ANT_MAIN Antenna Interfaces	53
5.7.2 ANT_GNSS Antenna Interface	55

5.7.3 ANT_WLAN_2G Antenna Interface	57
5.7.4 Antenna Assembling	57
6 Electrical Characteristics and Reliability	60
6.1 Electrical Characteristics	60
6.2 Temperature Characteristics	60
6.3 ESD Protection	61
7 RF Characteristics	62
7.1 Operating Band	62
7.2 TX Power and RX Sensitivity	62
7.3 N706 GNSS Technical Parameters	63
7.4 WLAN Technical Parameters	64
8 Mechanical Characteristics	65
8.1 Dimensions	65
8.2 Label	65
8.3 Packing	66
8.3.1 Reels	66
8.3.2 Moisture	67
9 Mounting	68
9.1 N706 Foot Print	68
9.2 Application Foot Print	69
9.3 Stencil	69
9.4 Solder Paste	69
9.5 SMT Oven Temperature Profile	70

Figures

Figure 2-1 Block diagram	13
Figure 4-1 N706 pin definition (top view)	16
Figure 5-1 Recommended design 1	25
Figure 5-2 Recommended design 2	26
Figure 5-3 Recommended design 3	27
Figure 5-4 Recommended design 4	27
Figure 5-5 Reference design of the VRTC power supply	29
Figure 5-6 Reference design of startup controlled by a button	31
Figure 5-7 Reference design of startup controlled by an MCU	31
Figure 5-8 Startup process	32
Figure 5-9 Reference design of automatic start once powered up	32
Figure 5-10 N706 hard shutdown process	33
Figure 5-11 Resetting the module by using a button	34
Figure 5-12 Reset circuit with triode separating	34
Figure 5-13 Reset process of the module	35
Figure 5-14 Process of entering sleep mode	36
Figure 5-15 Incoming call service process	36
Figure 5-16 Outgoing call service process	37
Figure 5-17 Process of existing from the sleep mode	37
Figure 5-18 Reference design of module sleep and wake-up control	38
Figure 5-19 Recommended design of the USB interface	39
Figure 5-20 Recommended design of the UART interface	41
Figure 5-21 Recommended level shifting circuit1	41
Figure 5-22 Recommended level shifting circuit 2	42
Figure 5-23 Recommended level shifting circuit 3	43
Figure 5-24 Reference design of the USIM card (with hot-swap) interface	45
Figure 5-25 Reference design of the USIM card (without hot-swap) interface	46

Figure 5-26 Recommended design of the SPI interface	47
Figure 5-27 Reference design of the I2C interface	47
Figure 5-28 Reference design of the I2S interface	48
Figure 5-29 PCM reference design	49
Figure 5-30 Reference design of USB force download	51
Figure 5-31 Driving LED indicator with a triode	52
Figure 5-32 Driving LED indicator with a triode	52
Figure 5-33 L-type network	53
Figure 5-34 T-type network	54
Figure 5-35 π -type network.....	54
Figure 5-36 Recommended RF PCB design	55
Figure 5-37 GNSS RF structure.....	55
Figure 5-38 Reference design of passive GNSS antenna.....	56
Figure 5-39 Reference design of the active GNSS antenna.....	56
Figure 5-40 Murata RF connector encapsulation specifications.....	58
Figure 5-41 RF cable connections	58
Figure 5-42 Antenna layout	59
Figure 5-43 Layout around the antenna.....	59
Figure 8-1 Top and side view dimensions (unit: mm).....	65
Figure 8-2 N706 label.....	66
Figure 9-1 N706 foot print (bottom view, unit: mm).....	68
Figure 9-2 N706 PCB foot print (top view, unit: mm)	69
Figure 9-3 Oven temperature profile	70

Tables

Table 2-1 Variants and frequency bands.....	12
Table 2-2 Interface differences among different variants	12
Table 4-1 IO definitions and DC characteristics	17
Table 4-2 Pin descriptions	18
Table 5-1 Module startup descriptions	30
Table 5-2 LCD interface pinouts.....	49
Table 5-3 Camera interface pinouts	50
Table 6-1 N706 operating conditions	60
Table 6-2 Current consumption of N706 (typical).....	60
Table 6-3 ESD protection	61
Table 7-1 N706 operating bands	62
Table 7-2 N706 RF transmit power	62
Table 7-3 RF RX sensitivity of N706	63
Table 7-4 GNSS technical parameters.....	63
Table 7-5 WLAN TX power and RX sensitivity.....	64

About This Document

Scope

This document is applicable to the N706 series.

Audience

This document is intended for system engineers (SEs), development engineers, and test engineers.

Change History

Issue	Date	Change	Changed By
1.0	2022-04	Initial release.	Yan Xique
1.1	2022-06	- Added new N706 variant. - Added Table 2-2 "Interface differences among different variants". - Modified the pin name of PIN19, PIN25, PIN38, PIN39, PIN59, and PIN60. - Modified the voltage detection range of the ADC interface. - Updated Figure 9-1 "N706 foot print (bottom view, unit: mm)" and Figure 9-2 "N706 PCB foot print (top view, unit: mm)".	Yan Xique
1.2	2022-07	Changed the typical value of VBAT to 3.6 V.	Yan Xique

Conventions

Symbol	Indication
	Indicates danger or warning. This information must be followed. Otherwise, a catastrophic module or user device failure or bodily injury may occur.
	Indicates caution. This symbol alerts the user to important points about using the module. If these points are not followed, the module or user device may fail.
	Indicates instructions or tips. This symbol provides advices or suggestions that may be useful when using the module.

Related Documents

Neoway_N706_Datasheet

Neoway_N706_Product_Specifications

Neoway_N706_AT_Commands_Mannual

Neoway_N706_EVK_User_Guide

Neoway Confidential

1 Safety Recommendations

Ensure that this product is used in compliance with the requirements of the country and the environment. Please read the following safety recommendations to avoid body hurts or damages of product or workplace:

- Do not use this product at any places with a risk of fire or explosion such as gasoline stations, oil refineries, and so on.

If the product is used in a place with flammable gas or dust such as propane gas, gasoline, or flammable spray, the product will cause an explosion or fire.

- Do not use this product in environments such as hospital or airplane where it might interfere with other electronic equipment.

If the product is used in medical institutions or on airplanes, electromagnetic waves emitted by this product may interfere with surrounding equipment.

Follow the requirements below in design and use of the application for this module:

- Do not disassemble the module without permission from Neoway. Otherwise, we are entitled to refuse to provide further warranty.
- Design your application correctly by referring to the HW design guide document and our review feedback on your PCB design. Connect the product to a stable power supply and lay out traces following fire safety standards.
- Please avoid touching the pins of the module directly in case of damages caused by ESD.
- Do not insert/remove a USIM card or memory card in the module while it is not powered off.

2 About N706

N706 is an LTE industrial-grade cellular module developed based on Unisoc UIS8850, supporting FDD-LTE and TDD-LTE communications. N706 has dimensions of (17.7 ± 0.10) mm $\times$ (15.8 ± 0.10) mm $\times$ (2.5 ± 0.15) mm and supports rich hardware interfaces including audio, video, Wi-Fi scan and GNSS (optional). It is suitable for developing IoT communication devices such as wireless meter reading terminals, vehicle/handheld POS, industrial routers and so on.

N706 has the following characteristics:

- 500 MHz Cortex-A5 application processor with 32 KB I-cache and 32 KB D-cache
- Supported network mode: LTE Cat.1
- Supported interfaces: I2S/PCM, USIM, UART, I2C, USB2.0, GPIO, SPI, and ADC.

2.1 Product Overview

The following table lists the variants and frequency bands that N706 supports.

Table 2-1 Variants and frequency bands

Variant	Region	Band	GNSS ¹⁾	Memory
CA-01	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Supported	RAM: 128 Mb ROM: 64 Mb
CA-02	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Not supported	RAM: 64 Mb ROM: 32 Mb
CA-04-P1	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B38, B39, B40, B41	Not supported	RAM: 64 Mb ROM: 32 Mb

Table 2-2 Interface differences among different variants

Version	I2S interfaces	SPI (Camera)	SPI (LCD)	USIM1
CA-01	Supported	Supported	Supported	Supported
CA-02	Not supported	Not supported	Not supported	Not supported
CA-04-P1	Not supported	Not supported	Not supported	Not supported



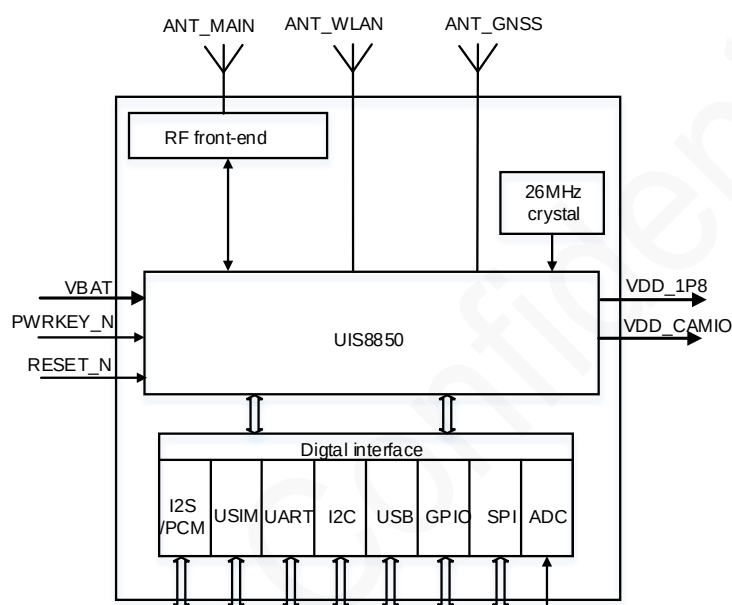
GNSS¹⁾ is optional.

2.2 Block Diagram

N706 consists of the following functionality units:

- Baseband
- Digital interfaces (I2S/PCM, USIM, UART, I2C, USB2.0, GPIO, SPI)
- Analog interface (ADC)

Figure 2-1 Block diagram



2.3 Basic Features

Parameter	Description
Physical features	• Dimensions: (17.7±0.10) mm × (15.8±0.10) mm × (2.3±0.15) mm • Package: 94-pin LGA • Weight: TBD
Temperature ranges	• Operating: -30°C to +75°C • Extended*: -40°C* to +85°C* • Storage: -40°C to +90°C
Operating voltage	VBAT: 3.4 V- 4.2 V, typical value: 3.6 V
Operating current	Sleep mode ²⁾ : TBD Standby mode ³⁾ : TBD Operating mode ⁴⁾ : TBD

Application processor	ARM Cortex-A5 processor, main frequency 500 MHz.
Band	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B38, B39, B40, B41
Wireless rate	FDD-LTE: Cat.1, Max 10 Mbps (DL)/Max 5 Mbps (UL) TDD-LTE: Cat.1, Max 8 Mbps (DL)/Max 2 Mbps (UL)
Application interfaces	4G antenna, GNSS antenna, and WLAN antenna. All of them have a characteristic impedance of 50 Ω .
	Three UART interfaces, with a maximum speed of 115200 bps. • UART 1 is used for AT communications. • UART 2 is used to capture CP-side log. • UART 4 is used to capture AP-side log.
	Two USIM interfaces, 1.8 V/3 V adaptive
	One USB 2.0 interface, supporting only slave mode.
	Three SPI interfaces One standard SPI interface supports only master mode. • One LCD-dedicated SPI interface and can be multiplexed as Flash SPI. • One camera-dedicated SPI interface
	Two 12-bit ADC interfaces, can detect voltages ranging from 0.1 V to VBAT.
	Two I2C interfaces, master mode only.
AT commands	One I2S interface, compatible with the PCM interface.
	3GPP Release 13 Neoway extended commands
SMS	PDU, TXT
Data	PPP, RNDIS, ECM
Protocol	TCP, UDP, MQTT, FTP, HTTP/HTTPS, SSL, TLS
Certification approval	CCC*, SRRC*, RoHS*, CE*, CTA*

* indicates in development.



"Sleep mode²⁾": the module enters a low power consumption state. In this state, the peripheral interface of the module is disabled, but the radio frequency (RF) is functioning properly. The module will exit the sleep mode when there is an incoming call or SMS message, and will re-enter the sleep mode at the end of the incoming call or conversation.

"Standby mode³⁾": the module is in normal working state, but there is no on-going data service.

"Operating mode⁴⁾": the current of the module when data communication is available.

3 Reference Standards

N706 is designed by referring to the following standards:

- 3GPP TS 36.521-1 V13.0.0 User Equipment(UE) conformance specification; Radio transmission and reception; Part 1: Conformance Testing
- 3GPP TS 21.111 V13.0.0 USIM and IC card requirements
- 3GPP TS 51.011 V4.15.0 Specification of the Subscriber Identity Module -Mobile Equipment (SIM-ME) interface
- 3GPP TS 31.102 V13.0.0 Characteristics of the Universal Subscriber Identity Module (USIM) application
- 3GPP TS 31.111 V13.0.0 Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)
- 3GPP TS 27.007 V13.0.0 AT command set for User Equipment (UE)
- 3GPP TS 27.005 V13.0.0 Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE - DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)

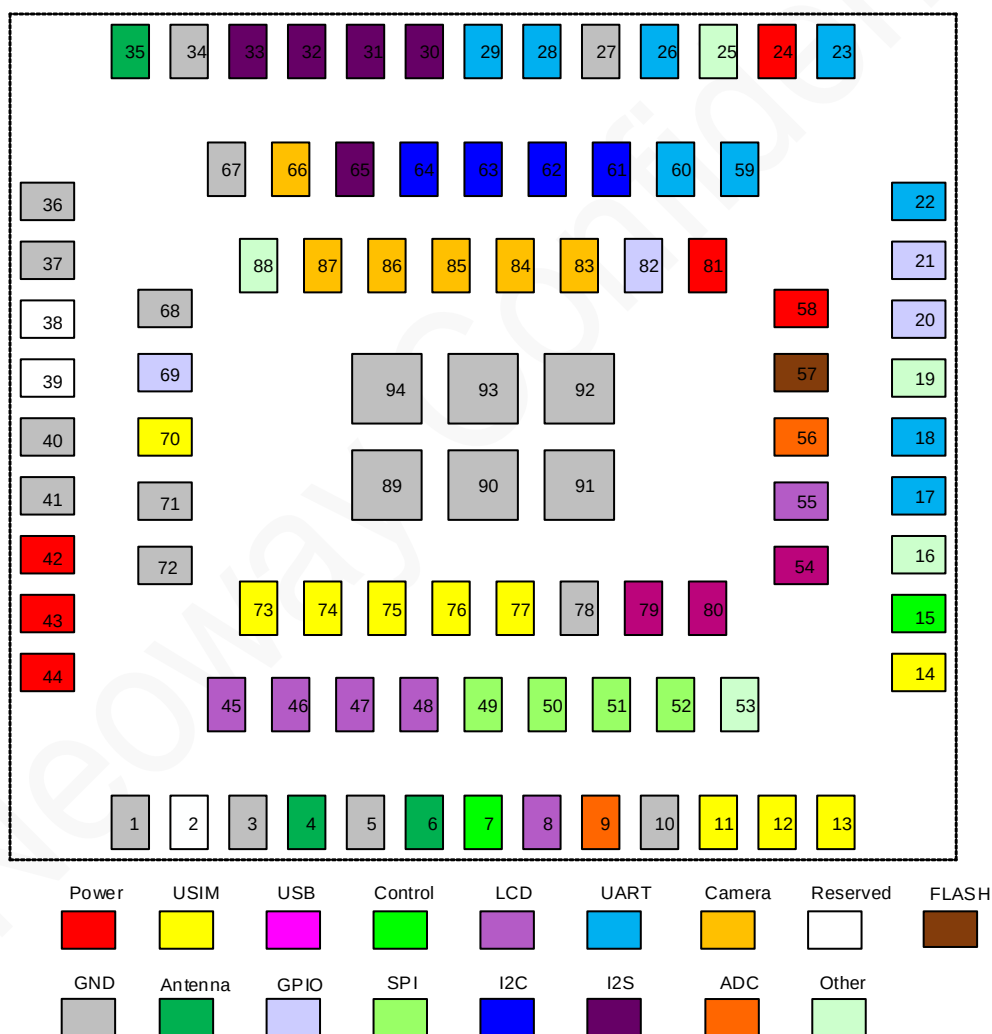
4 Module Pins

There are 94 pins on N706 and their pads are introduced in LGA package.

4.1 Pin Layout

The following figure shows the pad layout of N706.

Figure 4-1 N706 pin definition (top view)



4.2 Pin Descriptions

The following table lists the IO definitions and DC characteristics.

Table 4-1 IO definitions and DC characteristics

IO type			
B	Digital input/output, CMOS logic level		
DO	Digital output, CMOS logic level		
DI	Digital input, CMOS logic level		
PO	Power output		
PI	Power input		
AO	Analog output		
AI	Analog input		
AIO	Analog input/output		
Level feature			
Interface type	Power domain	Logic level descriptions	
USIM	P1: 1.8 V/3 V self-adaptive	1.8 V level feature: $V_{IH}=0.7 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P1}$ $V_{OH}=0.9 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P1}$	3 V level feature: $V_{IH}=0.7 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P1}$ $V_{OH}=0.9 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P1}$
GPIO	P2:1.8V	$V_{IH}=0.7 \times V_{DD_P2} \sim V_{DD_P2}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P2}$ $V_{OH}=0.9 \times V_{DD_P2} \sim V_{DD_P2}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P2}$	

Table 4-2 Pin descriptions

Signal	Pin	I/O type	Function description	Level feature	Remarks
Power interfaces					
VBAT	42, 43	PI	Main power supply of the module	$V_{\min} = 3.4 \text{ V}$ $V_{\text{norm}} = 3.6 \text{ V}$ $V_{\max} = 4.2 \text{ V}$	The external power supplies at least 1.5 A current to VBAT.
VDD_1P8	24	PO	1.8 V power output	$V_{\text{norm}} = 1.8 \text{ V}$ $I_{\max} = 50 \text{ mA}$	IO level, used only for level shifting. Leave this pin floating if it is not used.
VRTC	44	PI	RTC power supply	2.8 V - 3.2 V, typical value: 3.0 V	Supplies power for the real-time clock inside the module. It can be externally connected to button cells or Farah capacitors.
GND	1, 3, 5, 10, 27, 34, 36, 37, 40, 41, 67, 68, 71, 72, 78, 89-94				Ensure that all GND pins are grounded.
Control interfaces					
PWRKEY_N	7	DI	ON/OFF button	-	Triggered by a low level; used to control startup/shutdown of the module. This pin is connected to VBAT through a 20 k Ω pull-up resistor inside the module.
RESET_N	15	DI	Module reset input	-	Triggered by a low level to reset the module. This pin is connected to VBAT through a 20 k Ω pull-up resistor inside the module.
SLEEP/WAKEUP	19	DI	Sleep mode control/PSM wakeup	P2	See section 5.2.4 "SLEEP/Wakeup".
UART interfaces					
UART1_RXD	17	DI	Data receiving	P2	Standard version: used for AT communications Open version: used for data transmission.
UART1_TXD	18	DO	Data transmitting	P2	
UART1_CTS	22	DI	The user allows the module to send	P2	

			data.		
UART1_RTS	23	DO	The module requests the user to send data.	P2	UART2 is used to capture CP-side log.
UART2_RXD	28	DI	Data receiving	P2	
UART2_TXD	29	DO	Data transmitting	P2	
UART2_CTS	26	DI	The user allows the module to send data.	P2	
DEBUG_UART4_RXD	60	DI	Data receiving	P2	UART4 is use for debugging. It can be used to capture AP-side log.
DEBUG_UART4_TXD	59	DO	Data transmitting	P2	
USIM interfaces					
USIM0_DATA	11	B	USIM0 data	P1	Connecting a 4.7 kΩ pull-up resistor to USIM0_VCC is required.
USIM0_RESET	12	DO	USIM0 reset	P1	-
USIM0_CLK	13	DO	USIM0 clock	P1	-
USIM0_VCC	14	PO	USIM0 power supply	P1	-
USIM0_DET	77	DI	USIM0 detection	P2	If the hot-swapping function is not used, see 5.3.3 “USIM”.
USIM1_VCC	70	PO	USIM1 power supply	P1	-
USIM1_RESET	73	DO	USIM1 reset	P1	-
USIM1_CLK	74	DO	USIM1 clock	P1	-
USIM1_DATA	75	B	USIM1 data	P1	Connecting a 4.7 kΩ pull-up resistor to USIM1_VCC is required.

USIM1_DET	76	DI	USIM1 detection	P2	If the hot-swapping function is not used, see 5.3.3 "USIM."
USB interface					
USB_VBUS	54	PI	USB power supply	4.75 V - 5.25 V, typical value: 5 V	-
USB_DP	79	AIO	USB data positive signal	-	Route DM and DP traces in differential mode, control impedance of 90 Ω.
USB_DM	80	AIO	USB data negative signal	-	
ADC interfaces					
ADC1	9	AI	Analog-to-digital signal	-	12-bit, can detect voltage ranging from 0.1 V to VBAT. Leave this pin floating if it is not used.
ADC2	56	AI	Analog-to-digital signal	-	12-bit, can detect voltage ranging from 0.1 V to VBAT. Leave this pin floating if it is not used.
I2C interfaces					
I2C1_SDA	61	B	I2C data	P2	Connecting this pin to VDD_1P8 through a pull-up resistor is required.
I2C1_SCL	62	DO	I2C clock	P2	Connecting this pin to VDD_1P8 through a pull-up resistor is required.
I2C2_SDA	63	B	I2C data	P2	Connecting this pin to VDD_1P8 through a pull-up resistor is required.
I2C2_SCL	64	DO	I2C clock	P2	Connecting this pin to VDD_1P8 through a pull-up resistor is required.
I2S interfaces					
I2S_SCLK	30	DO	I2S data clock	P2	Can be multiplexed as PCM_SCLK.

I2S_WS	31	DO	I2S right and right channels selection	P2	Can be multiplexed as PCM_SYNC.
I2S_RX	32	DI	I2S data receiving	P2	Can be multiplexed as PCM_DIN.
I2S_TX	33	DO	I2S data sending	P2	Can be multiplexed as PCM_DOUT.
I2S_MCLK	65	DO	I2S master clock	P2	-
LCD interface					
LCD_SPI_MOSI	45	DO	SPI input of the slave device and output of the master device	P2	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_CS.
LCD_SPI_MISO	46	DI	SPI output of the slave device and input of the master device	P2	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_CLK.
LCD_SPI_CS	47	DO	SPI chip select signal of the slave device	P2	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_MISO.
LCD_SPI_CLK	48	DO	SPI clock	P2	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_MOSI.
LCD_TE	8	DO	LCD frame synchronization	P2	LCD-dedicated SPI interface Can be multiplexed as FLASH_HOLD.
LCD_RST_N	55	DO	LCD reset	P2	-
SPI interface					
SPI2_MISO	49	DI	SPI output of the slave device and input of the master device	P2	Leave this pin floating if it is not used.
SPI2_MOSI	50	DO	SPI input of the slave device and output of the master device	P2	Leave this pin floating if it is not used.
SPI2_CS	51	DO	SPI chip select signal of the slave device	P2	Leave this pin floating if it is not used.

SPI2_CLK	52	DO	SPI clock	P2	Leave this pin floating if it is not used.
CAM interfaces					
AVDD_CAM	58	PO	CAM analog power	V _{min} = 1.61 V V _{norm} = 1.8 V V _{max} = 3.2 V	The power supply is disabled by default. After it is enabled, it outputs 1.8 V/100 mA by default.
CAM_PWDN	66	DO	Power down control	P2	Leave this pin floating if it is not used.
VDD_CAMIO	81	PO	CAM IO power	V _{min} = 1.4 V V _{norm} = 1.8 V V _{max} = 1.85 V	The power supply is disabled by default. After it is enabled, it outputs 1.8 V/100 mA by default.
CAM_SPI_MISO	83	DI	SPI output of the slave device and input of the master device	P2	Camera-dedicated SPI interface
CAM_SPI_MOSI	84	DO	SPI input of the slave device and output of the master device	P2	
CAM_SPI_CLK	85	DO	SPI clock	P2	
CAM_RST_N	86	DO	SPI reset	P2	-
CAM_MCLK	87	DO	CAM main clock	P2	-
GPIO interfaces					
GPIO_1	20	B	GPIO	P2	During the startup process, the GPIO level is uncertain, a pull-up or pull-down resistor should be added according to different usage scenarios.
GPIO_2	21	B	GPIO	P2	
GPIO_3	69	B	GPIO	P2	
GPIO_0	82	B	GPIO	P2	
Antenna interfaces					
ANT_GNSS	4	AI	GNSS antenna	-	50 Ω impedance characteristics

ANT_WLAN_2G	6	AI	2.4 GHz WLAN antenna	-	50 Ω impedance characteristics
ANT_MAIN	35	AIO	Main antenna	-	50 Ω impedance characteristics
Other interfaces					
NET_LIGHT	16	DO	Network status indication	P2	Leave this pin floating if it is not used.
RING	25	DO	Incoming call indicator control	P2	Leave this pin floating if it is not used.
USB_BOOT	53	DI	Forcible download control pin	P2	Ground this pin and then power up the module to enter the USB forced download mode.
FLASH_WP	57	B	FLASH write protection	P2	Leave this pin floating if it is not used.
GNSS_LNA_EN	88	DO	GNSS LNA enable	P2	Leave this pin floating if it is not used.
RESERVED	2, 38, 39	Reserved	-	-	Used for function extension or the function not open. These RESERVE pins might have different functions. Ensure that all RESERVED pins are left floating.

5 Application Interfaces

N706 provides the control, communications, peripherals, display, and RF interfaces to meet the functional requirements of customers in different application scenarios.

This chapter describes how to design each interface and provides reference designs and guidelines.

5.1 Power Interfaces

The schematic design and PCB layout of the power supply part are the most critical process in application design and they will determine the performance of customers' applications. Please read the design guidelines of power supply and comply with the correct design principles to obtain the optimal circuit performance.

Signal	Pin	I/O	Function description	Remarks
VBAT	42, 43	PI	Main power supply of the module	$V_{\min} = 3.4 \text{ V}$ $V_{\text{norm}} = 3.6 \text{ V}$ $V_{\max} = 4.2 \text{ V}$
VDD_1P8	24	PO	1.8 V power	$V_{\text{norm}} = 1.8 \text{ V}$ $I_{\max} = 50 \text{ mA}$
VRTC	44	PI	RTC power supply	2.8 V - 3.2 V, typical value: 3.0 V
GND	1, 3, 5, 10, 27, 34, 36, 37, 40, 41, 67, 68, 71, 72, 78, 89-94			Ensure that all GND pins are grounded.

5.1.1 VBAT

The power supply design covers two parts: schematic design and PCB layout.

Schematic Design



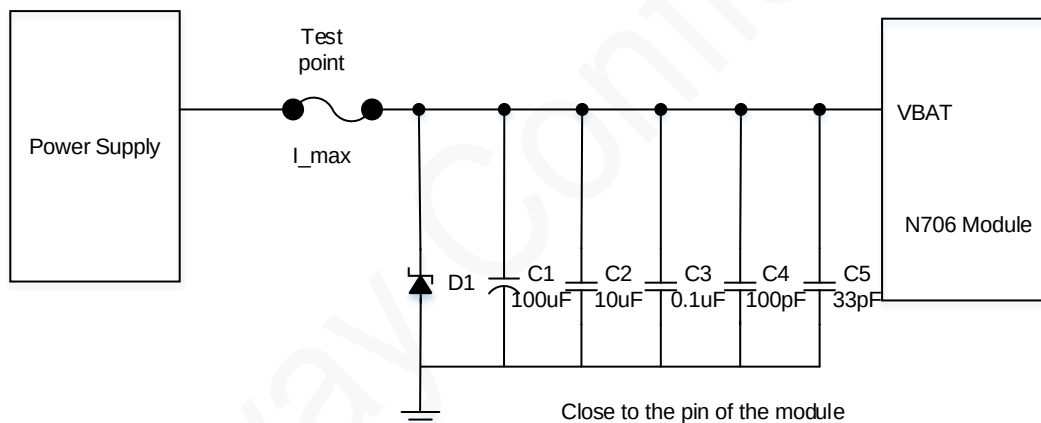
Never use a diode to make the drop voltage between a higher input and the module power supply. The forward voltage drop V_f of the diode has two characteristics: one is that it increases with the increase of forward current; the other is that it increases significantly at a low temperature. If there is an instantaneous high current, the instantaneous increase of forward voltage drop will lead to unstable operating voltage of the module, or even damage the module.

The power supply design of the N706 module is determined by the power input voltage. The designs are classified by power input voltage as follows:

- Supports the 3.4 V - 4.2 V power input (typical value: 3.6 V, using batteries for power supply)
- Supports the 4.2 V - 5.5 V power input (typical value: 5.0 V, using internal rectifiers of the computer for power supply)
- Supports the 5.5 V - 24 V power input (typical value: 12 V, typically applicable to the automobile industry)

The recommended 3.4 V to 4.2 V input design is as follows:

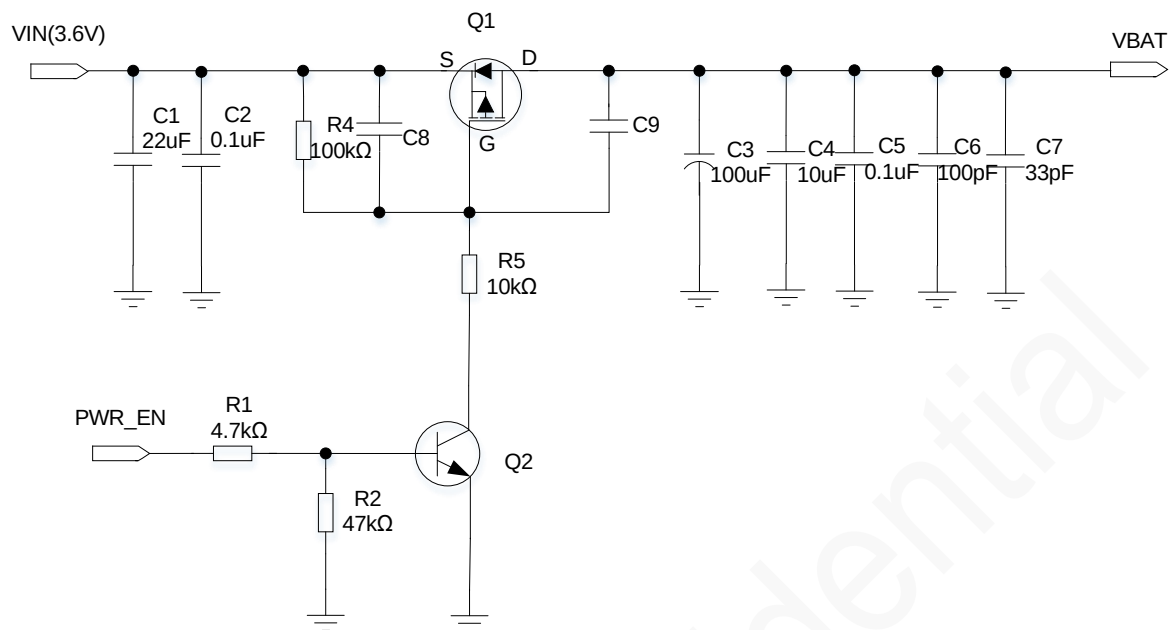
Figure 5-1 Recommended design 1



- The maximum input voltage for the module is 4.2 V and the typical value is 3.6 V.
- The reverse operating voltage (V_{RWM}) of the TVS diode (D1) is 4.5 V, and the peak power P_{pp} is 2800 W ($t_p=8/20$ uS). With surge protection function, place the TVS close to the input interface of the power input interface to clamp the surge voltage before it enters back-end circuits. Therefore, the back-end component and the module are protected.
- A large bypass tantalum capacitor (220 μ F or 100 μ F) or aluminum capacitor (470 μ F or 1000 μ F) is expected at C1 to reduce voltage drops during bursts. Its maximum safe operating voltage should be larger than 2 times the voltage across the power supply.
- Place a bypass capacitor (C2, C3, C4, C5) of low-ESR close to the module to filter out high-frequency jamming from the power supply.

The following circuit design is recommended to control the power supply.

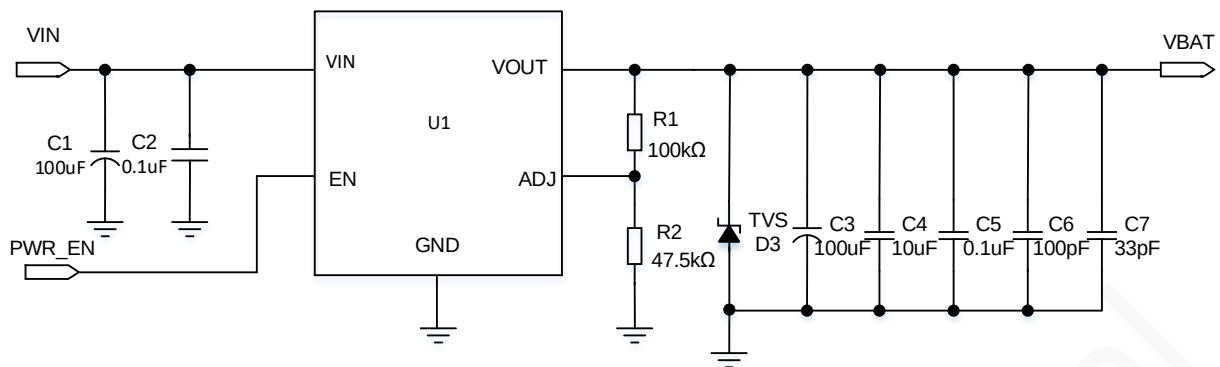
Figure 5-2 Recommended design 2



- Select an enhanced p-MOSFET at Q1, of which the safe operating voltage is at least 12 V ($V_{ds} = -12\text{ V}$) and drain current is at least 3.5 A ($I_{D(MAX)} = -3.5\text{ A}$) and R_{ds} is low ($R_{ds(on)} = 108\text{ m}\Omega$).
- Select a common NPN tripolar transistor at Q2. Reserve enough tolerances of R1 and R2 in design, especially for the situation in which operating voltage of the tripolar transistor might increase in low temperature; it is recommended that the value of R2 be at least 10 times that of R1.
- Place C3 close to the module. A large tantalum electrolytic capacitor (220 μF or 100 μF) or aluminum electrolytic capacitor (470 μF or 1000 μF) can be selected at C3 to improve the instantaneous large current freewheeling ability of the power supply. Its withstand voltage should be larger than 2 times the voltage of the power supply.
- Place a bypass capacitor (C4, C5, C6, C7) of low-ESR close to the module to filter out high-frequency jamming from the power supply.

Recommended 4.2 V to 5.5 V input design:

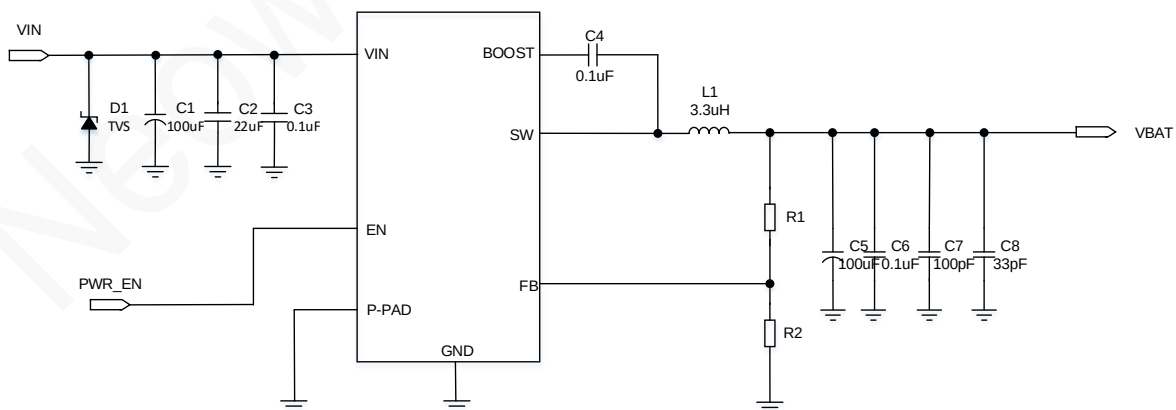
Figure 5-3 Recommended design 3



- Design with LDO is simpler and more efficient when the output of power supply is close to the permissible voltage across VBAT.
- Select an LDO that can output 1.5 A current at U1 to ensure the performance of the module.
- The reverse operating voltage (V_{RWM}) of the TVS diode (D3) is 4.5 V, and the peak power Ppp is 2800 W ($t_p=8/20$ uS). With surge protection function, place the TVS close to the input interface of the power input interface to clamp the surge voltage before it enters back-end circuits. Therefore, the back-end component and the module are protected.
- Place C3 close to the module. A large tantalum electrolytic capacitor (220 μ F or 100 μ F) or aluminum electrolytic capacitor (470 μ F or 1000 μ F) can be selected at C3 to improve the instantaneous large current freewheeling ability of the power supply. Its withstand voltage should be larger than 2 times the voltage of the power supply.
- Place the bypass capacitors (C4, C5, C6, C7) of low-ESR close to the module to filter out high-frequency jamming from the power supply.

Recommended 5.5 V to 24 V input design:

Figure 5-4 Recommended design 4



- As there is a big difference between the power input and VBAT, the DC-DC step-down chip should be selected, and the maximum output current should be at least 1.5 A.

- Place the TVS (D1) close to the power input interface to clamp the surge voltage before it enters back-end circuits. Therefore, the back-end components and the module are protected.
- Place C5 close to the module. A large tantalum electrolytic capacitor (220 μ F or 100 μ F) or aluminum electrolytic capacitor (470 μ F or 1000 μ F) can be selected at C5 to improve the instantaneous large current freewheeling ability of the power supply. Its withstand voltage should be larger than 2 times the voltage of the power supply.
- Place the bypass capacitors (C6, C7, C8) of low-ESR close to the module to filter out high-frequency jamming from the power supply.



When the module fails to shut down normally due to abnormality, it is recommended to cut off the power first to shut down the module, and then power on again to restart the module.

PCB Layout

A low ESR capacitor must be placed at the output end of the power supply to suppress the peak current. Place TVS diodes at the input end of the module to suppress voltage spikes and protect back-end devices. The circuit design is important, and the device layout and routing are equally important. Several key points in power supply design are summarized below:

- The TVS diode can absorb instantaneous high-power pulses, and withstand instantaneous pulse current peaks up to tens or even hundreds of amperes. The clamp response time is extremely short. Place the TVS as close as possible to the power input to ensure that the surge voltage can be clamped before the pulse is coupled to the adjacent PCB wires.
- Place the bypass capacitor as close as possible to the power supply pin of the module to filter out high-frequency noise signals in the power supply.
- For the module power circuit, ensure that the PCB trace width can pass a current of 1.5 A safely, and there should be no obvious loop voltage drop. Keep the PCB trace width be at least 1.5 mm and ensure that the ground plane of the power supply part is as complete as possible. In addition, try to make the power cable short and thick.
- Keep noise-sensitive circuits, such as audio circuits and RF circuits far away from power circuits, especially when the DC-DC power supply is used.
- The voltage frequency of the SW pin of the DC-DC power supply is high, and the loop should be minimized. Keep sensitive components far away from the SW pin of the DC-DC component to prevent noise coupling. Place feedback components as close as possible to the FB pin.
- Ground the GND pin and bottom pad of the chip to ensure good heat dissipation and noise isolation.

5.1.2 VDD_1P8



It is recommended that VDD_1P8 is used for level shifting only and an ESD protector should be added.

N706 provides one VDD_1P8 interface that supports 1.8 V voltage and up to 50 mA current.

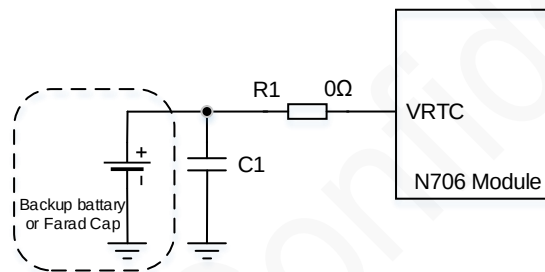
5.1.3 VRTC

The VRTC pin supplies power to the real-time clock inside the module.

When VBAT supplies power normally, the typical output voltage of V_{RTC} is 3.0 V and the maximum current is 2 mA. When VBAT is disconnected, connected it to a button cell or a Farad capacitor can maintain the normal operation of the RTC for a short time. Leave this pin floating if it is not used.

The following shows the reference design of the VRTC power supply.

Figure 5-5 Reference design of the VRTC power supply



5.2 Control Interfaces

Signal	Pin	I/O	Function description	Remarks
RESET_N	15	DI	Module reset input	Active low
PWRKEY_N	7	DI	ON/OFF button	Active low
SLEEP/WAKEUP	19	DI	Sleep mode control/PSM wakeup	The PSM wake-up function is in development.

5.2.1 Startup

Table 5-1 Module startup descriptions

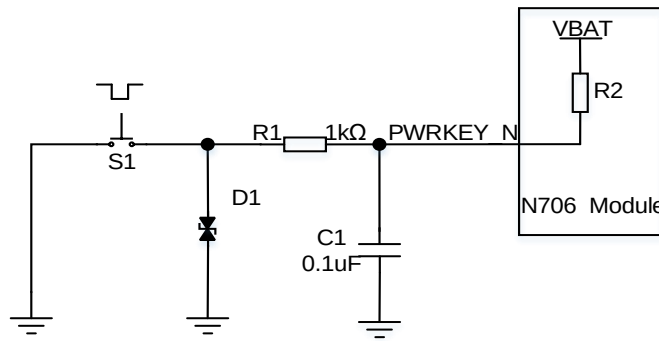
Startup method	Supported or not	Shutdown method	Cautions
Button	Supported	A long press of the power button for over 1.8s can shut down the module.	For example: When the module is working normally, do not directly disconnect the power supply of the module; otherwise, the flash memory (Flash) inside the module may be damaged.
Pulse	Supported	Inputting low pulse (controlled by MCUs) for not less than 1.8s can shut down the module.	
Automatic startup	Supported	Execute the shutdown AT command first and then power down the module.	For example: When using AT commands to shut down the module, if the system is not powered down, the module may start again after it is shut down.
USB startup	Supported	Unplug the USB, execute the shutdown AT command, and then power down the module.	For example: When using AT commands to shut down the module, if the system is not powered down and the USB is still connected, the module may start again after it is shut down.

Startup controlled by PWRKEY_N

The following show two recommended reference designs of startup controlled by PWRKEY_N:

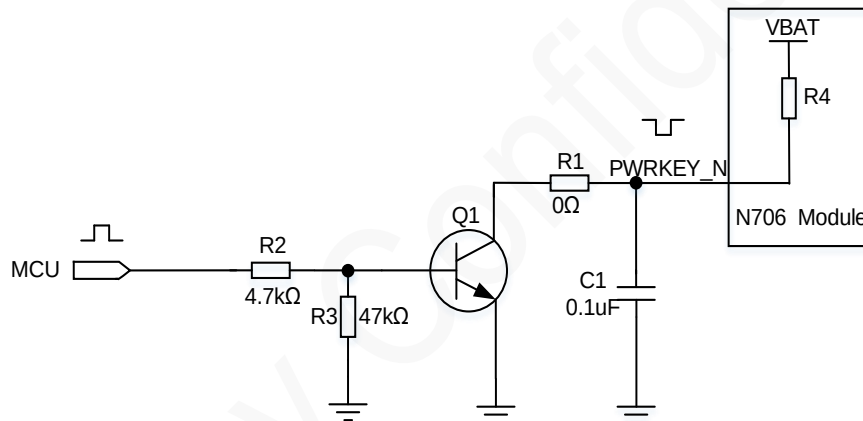
- Figure 5-6 shows the reference design of starting the module by using a button

Figure 5-6 Reference design of startup controlled by a button



- Figure 5-7 shows the reference design of starting the module through an MCU

Figure 5-7 Reference design of startup controlled by an MCU

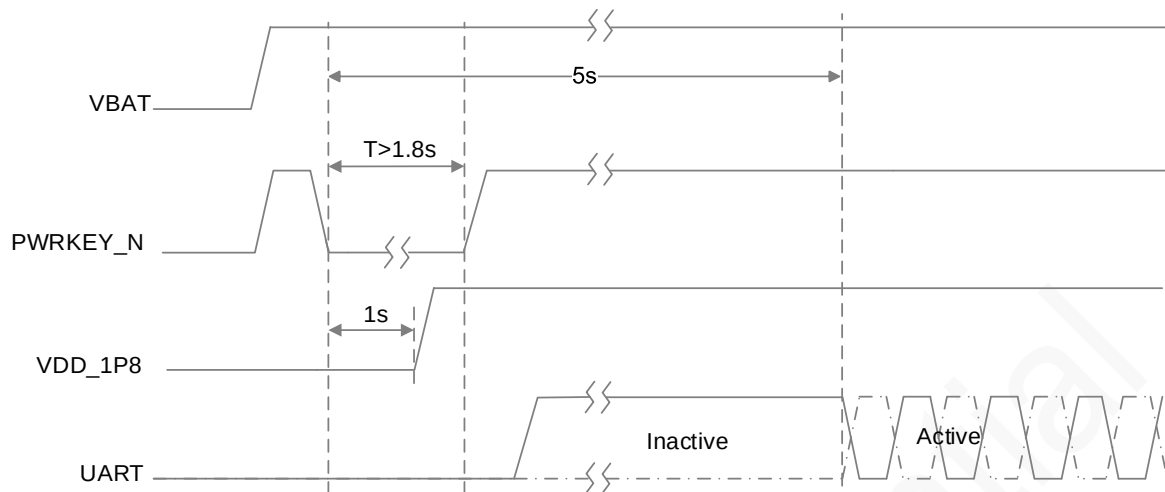


Startup process

To start the module, you can first power up VBAT, and then input a negative pulse for longer than 1.8s at PWRKEY_N. The AT function is available after 5 seconds. Do NOT connect an external resistor with large resistance in series to the PWRKEY_N pin (connecting a 1 kΩ resistor in series is recommended). Otherwise, the module cannot be started since the PWRKEY_N is at a high level all the time.

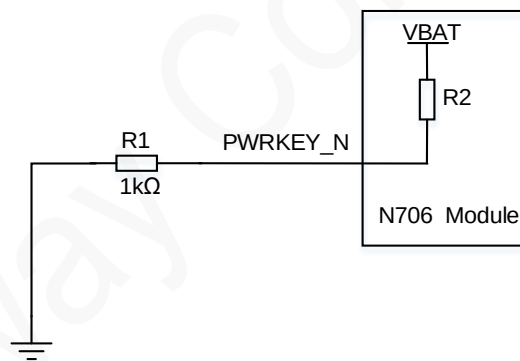
Do Not perform other operations on the module until it is initialized completely since the states of each pin are uncertain if the initialization process has not been completed. Figure 5-8 shows the startup process.

Figure 5-8 Startup process



Automatic start once powered up

Figure 5-9 Reference design of automatic start once powered up



If you implement the automatic startup function but the shutdown function is not required, you can directly pull up PWRKEY_N to ground. The recommended pull-down resistance is 1 kΩ.

5.2.2 Shutdown



- When the module is working normally, do not directly disconnect the power supply of the module; otherwise, the flash memory (Flash) inside the module may be damaged. It is recommended that before disconnecting the power supply, you should shut down the module through PWRKEY_N or AT-command shutdown.
- When shutting down the module by using the AT command, ensure that PWRKEY_N is at a high level after

the shutdown command is executed; otherwise, the module will start again after a shutdown.

Two methods are available to shut down the module: hard shutdown and soft shutdown.

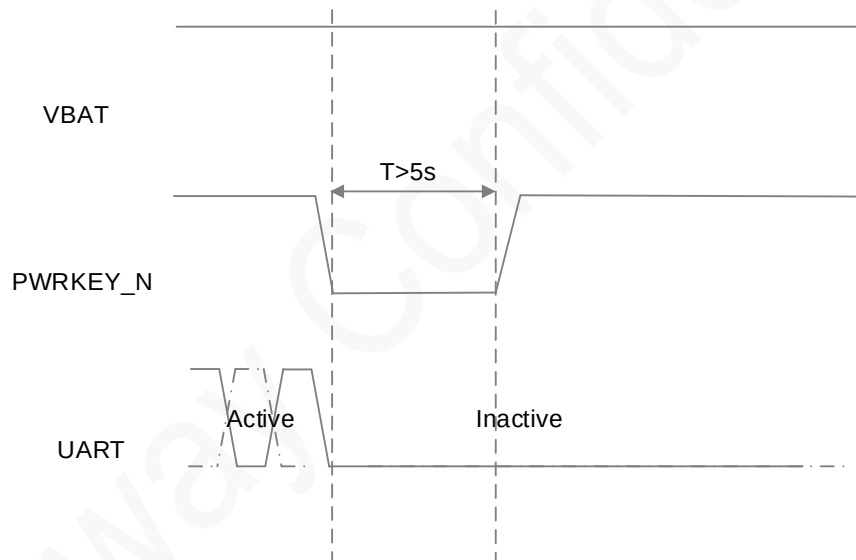
- Hard shutdown: shutdown controlled by PWRKEY_N
- Soft shutdown: shutdown by AT command execution

Hard shutdown

PWRKEY_N is used to hardware power off the module. When the module is working, inputting negative pulses for more than 5 seconds to PWRKEY_N can trigger the power-off state of the module.

The following figure shows the hardware shutdown process:

Figure 5-10 N706 hard shutdown process



Soft shutdown

When the module is on, sending the AT command can turn off the module. The shutdown state achieved by hard shutdown and soft shutdown is the same.

For details, see the *Neoway_N706_AT_Commands_Manual*.

5.2.3 Reset



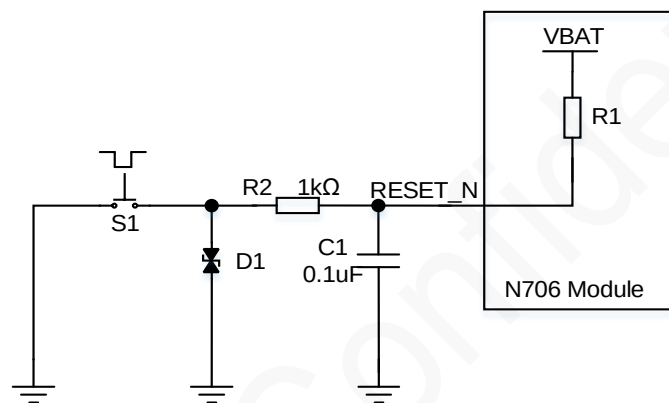
When the module encounters an exception (crash, etc.), you can reset the system through the RESET_N pin to ensure the normal operation of the module. Therefore, it is recommended to add a RESET_N control circuit during circuit design, and the recommended reference circuit is shown in Figure 5-11 and Figure 5-12.

RESET_N is used to reset the module. When the module is working, inputting a negative pulse for more than 50 ms to RESET_N can trigger the reset process of the module. Figure 5-11 shows the reset process.

The following show two recommended reference reset designs of N706:

- Resetting the module by using a button

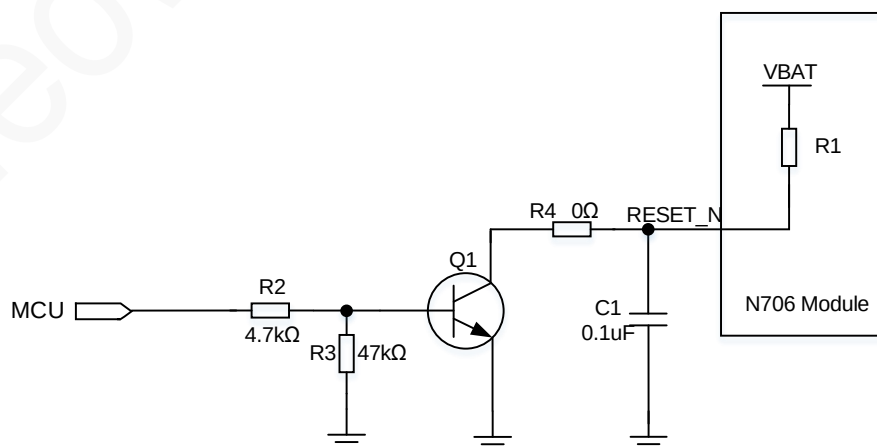
Figure 5-11 Resetting the module by using a button



- Resetting the module by controlling an MCU

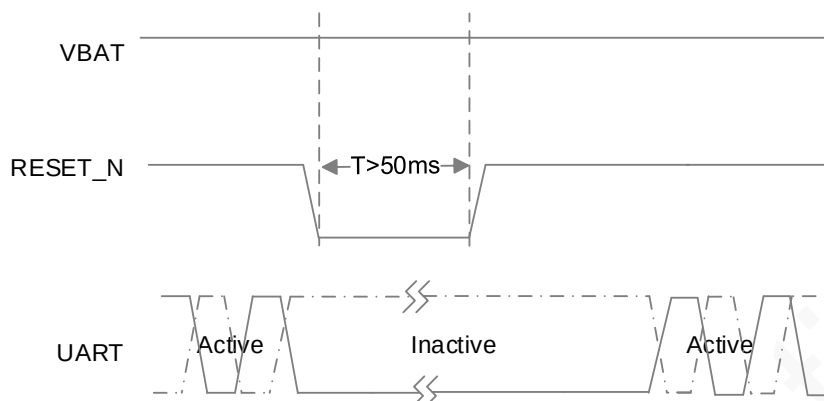
If you use a 1.8 V/2.8 V/3.0 V IO system, it is recommended to add a triode to separate it. Refer to the following designs.

Figure 5-12 Reset circuit with triode separating



The following figure shows the reset process of the N706 module:

Figure 5-13 Reset process of the module



5.2.4 SLEEP/WAKEUP

SLEEP

The SLEEP pin is used to control sleep mode of the module, and it needs to be used together with the AT commands. Entering sleep mode or waking up the module by executing AT commands through the USB interface is supported. For details, see the *Neoway_N706_AT_Commands_Manual*. In sleep mode, the module can also respond to incoming call, SMS, and data service in time.

The following show the processes:

Figure 5-14 Process of entering sleep mode

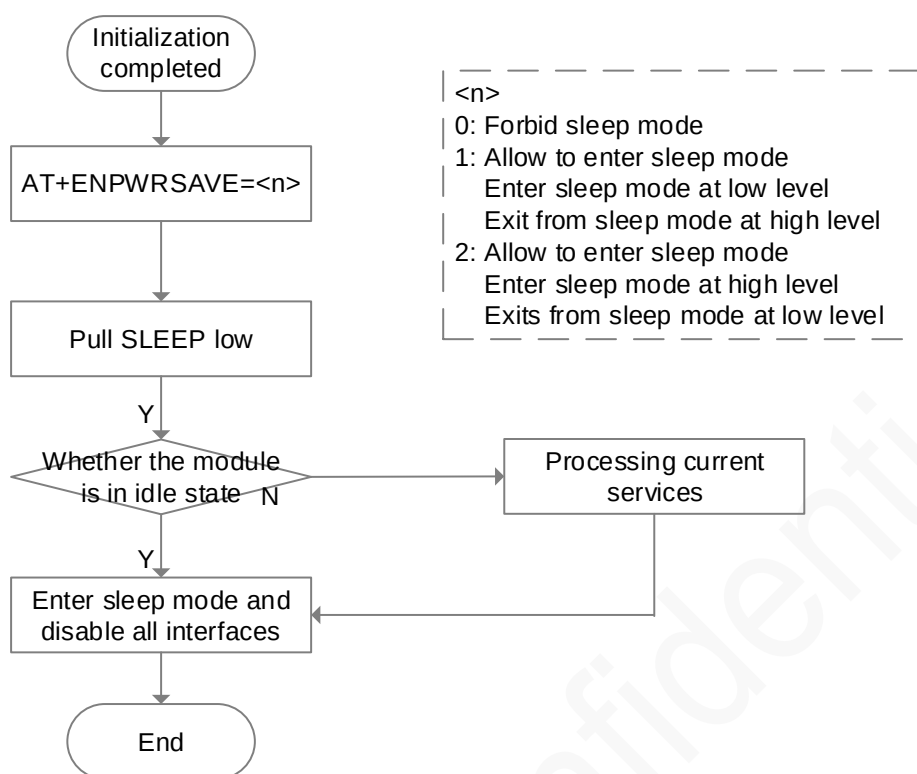


Figure 5-15 Incoming call service process

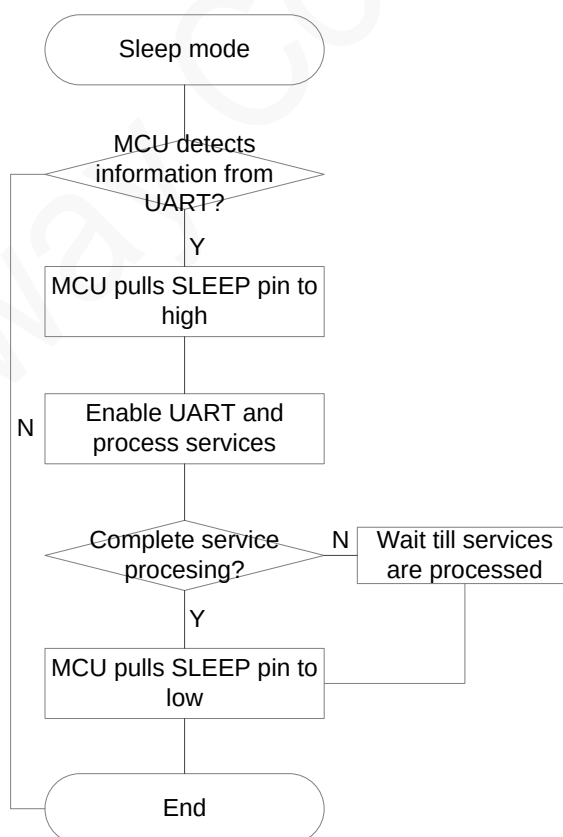


Figure 5-16 Outgoing call service process

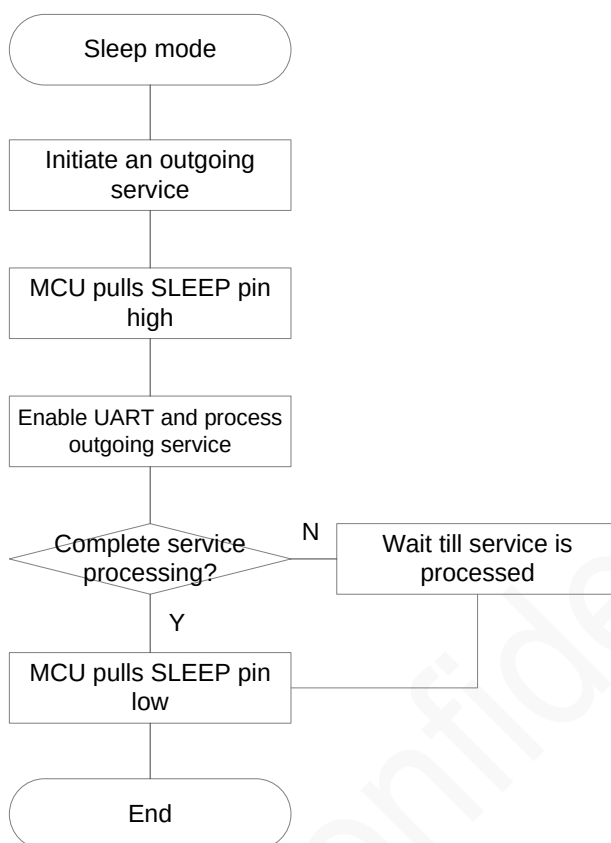
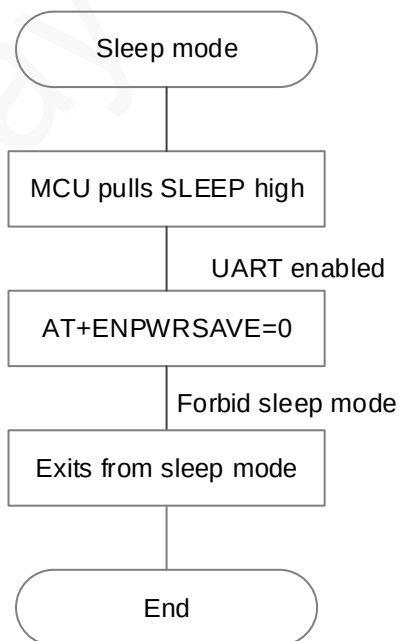


Figure 5-17 Process of existing from the sleep mode



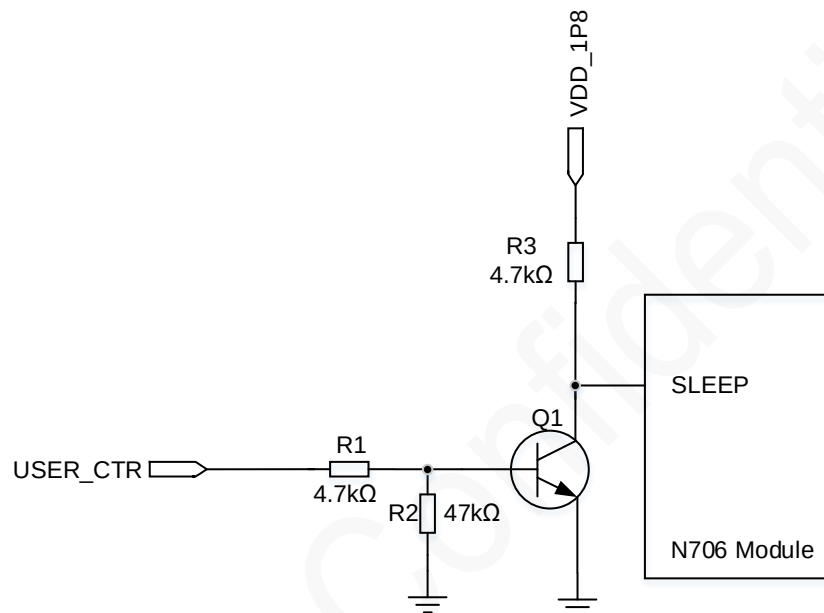
If you want to perform the wake-up function by using the sleep pin, it is recommended to design your

circuit by referring to Figure 5-18.

To enter into or exit from sleep mode, you should execute “AT+ENPWRSAVE=2” to allow sleep mode of the module, and then perform one of the following operations as required.

- Entering into sleep mode: keep USER_CTR low
- Exiting from sleep mode: keep USER_CTR high

Figure 5-18 Reference design of module sleep and wake-up control



WAKEUP



The wake-up function is in development.

5.3 Peripheral Interfaces

N706 provides various peripheral interfaces.

In all reference designs of this section, the receiving and sending directions included in the pin names of the peripheral interface of the module are based on the module, whereas peripheral pins are named based on the components. For example, UART_TXD indicates the pin used by the module to send data, and MCU_RXD indicates the pin used by the MCU to receive data. These two pins should be connected.

In the process of MCU selection and design, note whether the signal naming of pins is based on the

module or the MCU.

5.3.1 USB

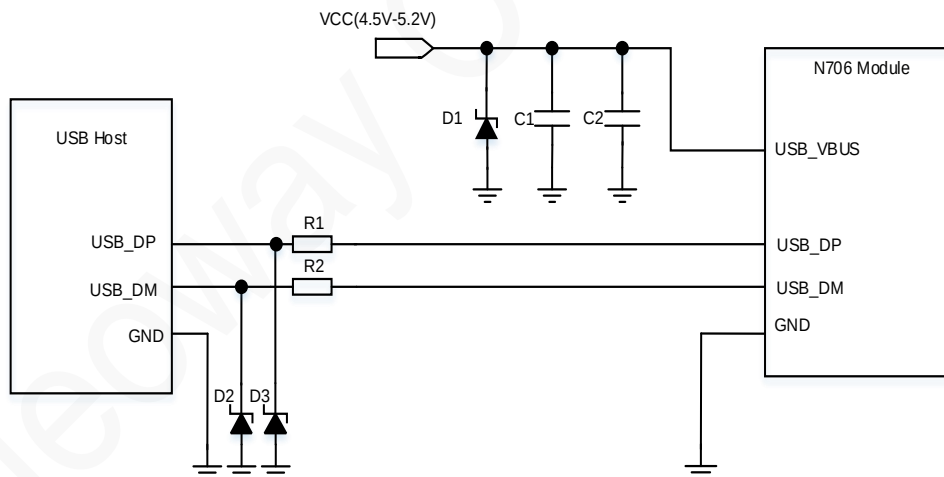


You should reserve the USB interface since it is used to debug module software.

Signal	Pin	I/O	Function description	Remarks
USB_VBUS	54	PI	USB power supply	4.75 V<USB_VBUS<5.25 V, typical value: 5 V
USB_DP	79	AIO	USB data positive signal	-
USB_DM	80	AIO	USB data negative signal	-

N706 can implement program download, data communications, and debugging through the USB interface. The USB of N706 only supports slave mode. Figure 5-19 shows the recommended USB connection circuit.

Figure 5-19 Recommended design of the USB interface



Schematic Design Guidelines

- Connect a 1 μF (C1) and a 33 pF (C2) filter capacitors in parallel to the USB_VBUS pin. An ESD component must be added for the power cable and the reverse working voltage V_{RWM} should be greater than 6 V.
- The junction capacitance of the ESD components D2 and D3 on the USB_DP and USB_DM cables must be smaller than 0.5 pF.

- Connecting a resistor less than 10 Ω in series to each of the USB_DP and USB_DM cables can effectively improve the ESD performance of the USB.

PCB Design Guidelines

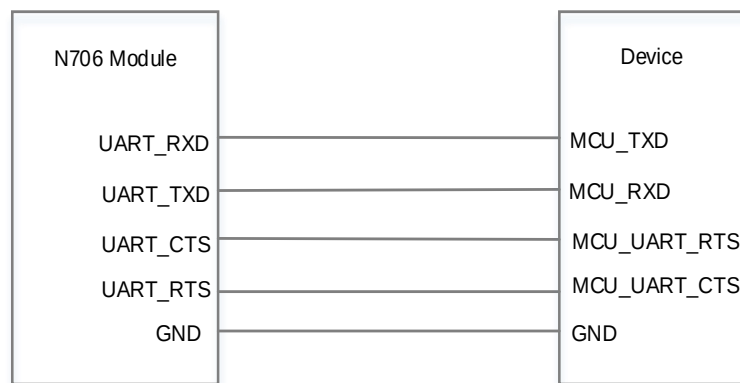
- Place the filter capacitor on the USB_VBUS as close as possible to the module pins, and place the ESD component as close as possible to the USB connector.
- Place the ESD component on the USB_DP and USB_DM cables as close as possible to the USB connector.
- USB data cables must adopt differential routing, and the differential impedance must be 90 Ω . The cable from the port to the module must be isolated from other signal cables and must be wrapped by ground cables.

5.3.2 UART

Signal	Pin	I/O	Function description	Remarks
UART1_RXD	17	DI	Data receiving	Standard version: used for AT communications Open version: used for data transmission.
UART1_TXD	18	DO	Data transmitting	
UART1_CTS	22	DI	The user allows the module to send data.	
UART1_RTS	23	DO	The module requests the user to send data.	
UART2_RXD	28	DI	Data receiving	UART2 is used to capture CP-side log.
UART2_TXD	29	DO	Data transmitting	
UART2_CTS	26	DI	The user allows the module to send data.	
DEBUG_UART4_RXD	60	DI	Data receiving	UART4 is use for debugging. It can be used to capture AP-side log.
DEBUG_UART4_TXD	59	DO	Data transmitting	

N706 provides three UART interfaces, which support 1.8 V level and baud rates up to 115200 bps. The following figure shows the recommended design of the UART interface.

Figure 5-20 Recommended design of the UART interface



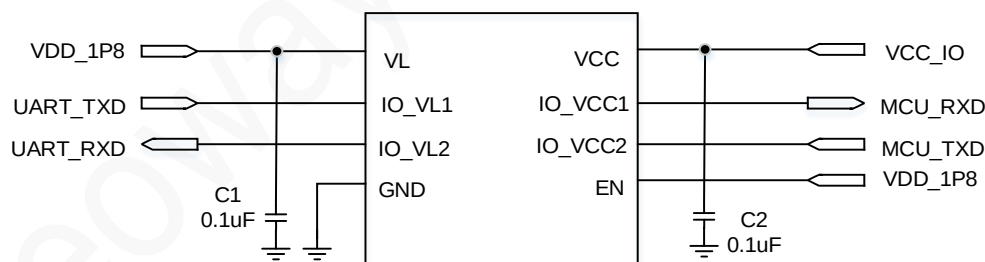
Schematic Design Guidelines

- Pay attention to the corresponding relations between the signal direction and the connection.
- It is prohibited to use diodes for level shifting.

If the UART does not match the logic voltage of the MCU, add a level shifting circuit outside of the module. Three level shifting circuits are recommended based on the differences in logic levels and rates.

- If the UART baud rate is greater than 115200 bps, it is recommended to refer to the recommended level shifting circuit 1. See Figure 5-21.

Figure 5-21 Recommended level shifting circuit1



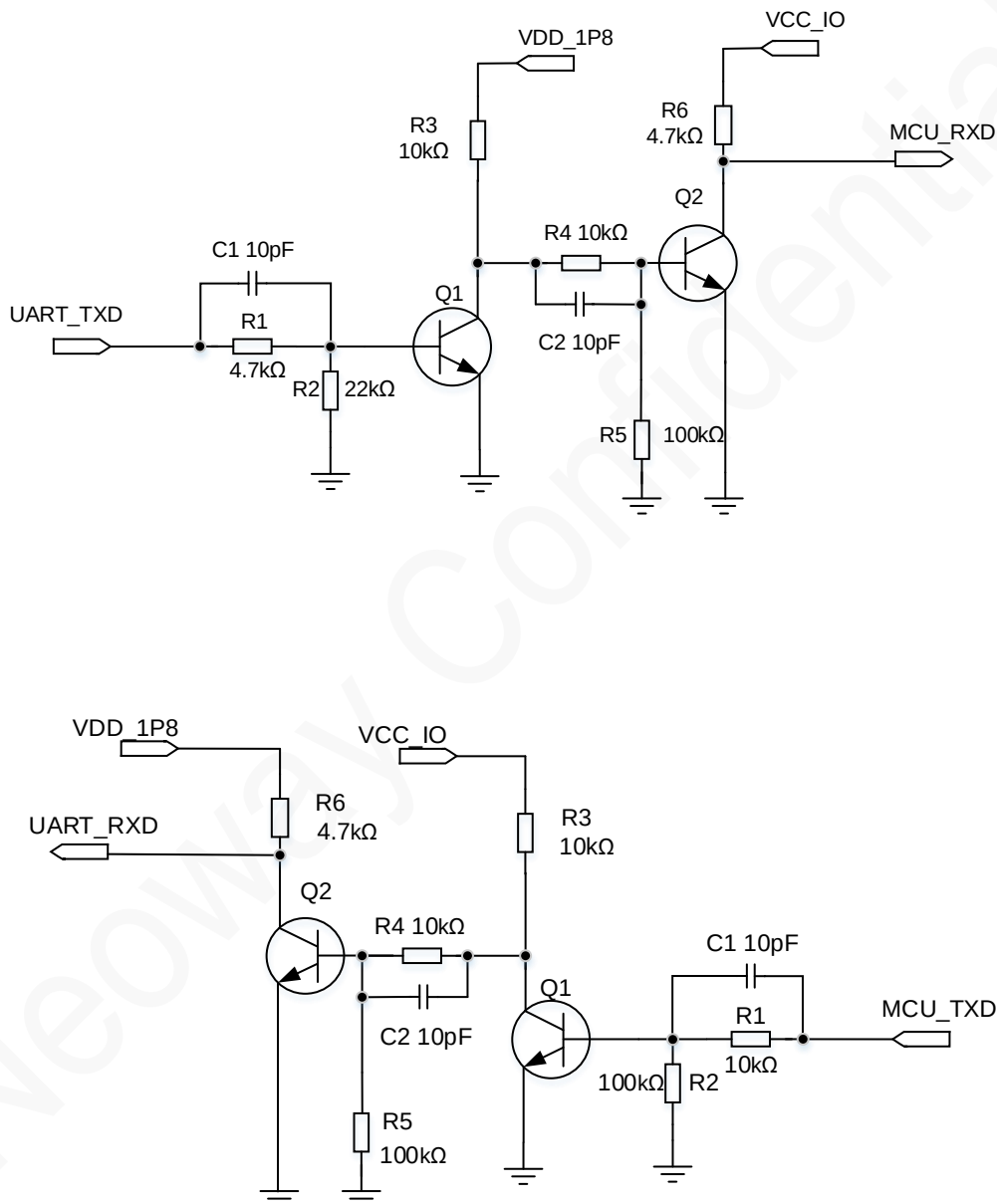
- VL is the reference voltage of IO_VL1 and IO_VL2, ranging from 1.5 V to 5.5 V.
- VCC is the reference voltage of IO_VCC1 and IO_VCC2, ranging from 1.5 V to 5.5 V.
- EN is the enable pin, which works at a voltage of greater than VL-0.2V. In the above circuit, the EN pin is connected to VDD_1P8 and the level shifter is always working.
- If the UART baud rate is less than or equal to 115200 bps, designing the serial ports TXD and RXD by referring to recommended level shifting circuit 2 and designing the serial ports CTS and

RTS by referring to recommended level shifting circuit 3 are recommended. As shown in Figure 5-22 and Figure 5-23.



Do Not directly apply the parameter values of the components provided by the circuit to your circuit design; they need to be adjusted according to the actual situations. Note the difference between different level shifting circuits.

Figure 5-22 Recommended level shifting circuit 2



MCU_TXD and MCU_RXD are the sending and receiving ports of the MCU respectively, and UART_TXD and UART_RXD are the sending and receiving ports of the module respectively. VCC_IO is the IO voltage of the MCU. VDD_1P8 is the IO voltage of the module.

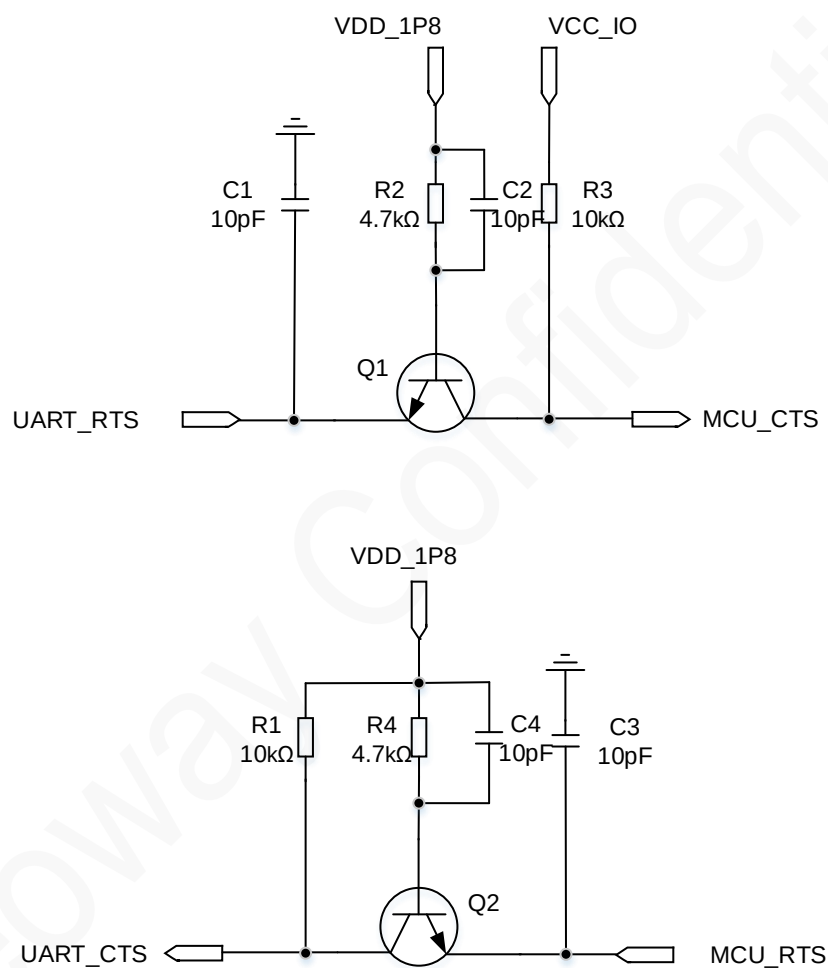
The circuit converts the voltage level through the turn-on and turn-off of the triode, and dual triode can

achieve higher pressure difference after level shifting.

Schematic Design Guidelines

- Ensure that the base voltage of the triode is operating within the temperature range and the transistor can be fully turned on.
- It is recommended to reserve the acceleration capacitor, which can adjust the delay of the level conversion circuit in some cases.

Figure 5-23 Recommended level shifting circuit 3



MCU_CTS and MCU_RTS are the MCU-side signals; UART_CTS and UART_RTS are the module-side signals. VCC_IO is the IO voltage of the MCU. VDD_1P8 is the IO voltage of the module.

The circuit convert the voltage level through the turn-on and turn-off of the triode, and the level shifting is unidirectional. Pay attention to the signal flow.

Schematic Design Guidelines

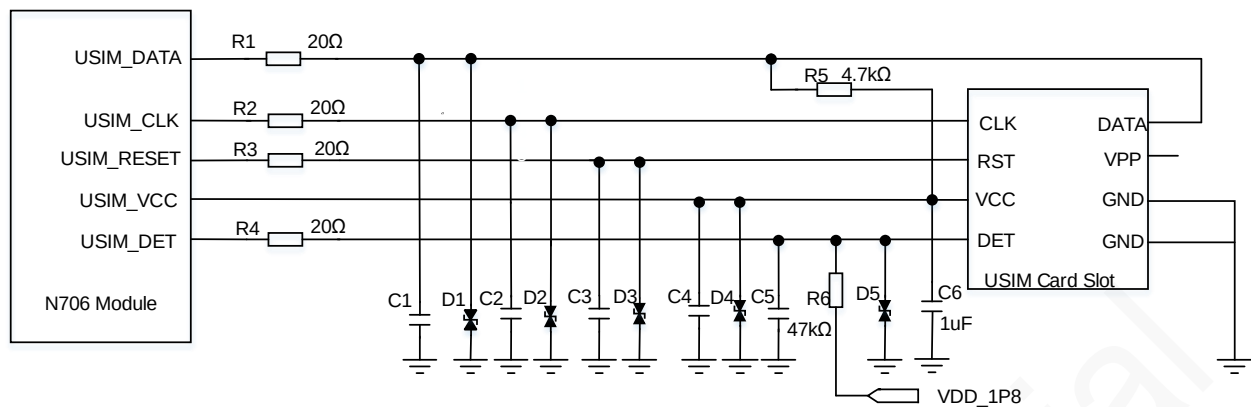
- Ensure that the voltage difference between the high-level and low-level sides is not greater than 2 V.
- For the accelerating capacitor, adjust it according to the actual test conditions; it is recommended to reserve the capacitor.
- The base voltage of the transistor is the lower voltage between both sides.

5.3.3 USIM

Signal	Pin	I/O	Function description	Remarks
USIM0_DATA	11	B	USIM0 data	Connecting a 4.7 kΩ pull-up resistor to USIM0_VCC is required.
USIM0_RESET	12	DO	USIM0 reset	
USIM0_CLK	13	DO	USIM0 clock	
USIM0_VCC	14	PO	USIM0 power supply	1.8 V/3.0 V (self-adaptive)
USIM0_DET	77	DI	USIM0 detection	The level of this interface is 1.8 V. If hot-swapping function is not used, pull the pin up to the VDD_1P8 with a 47 kΩ resistor.
USIM1_VCC	70	PO	USIM1 power supply	1.8 V/3.0 V (self-adaptive)
USIM1_RESET	73	DO	USIM1 reset	-
USIM1_CLK	74	DO	USIM1 clock	-
USIM1_DATA	75	B	USIM1 data	Connecting a 4.7 kΩ pull-up resistor to USIM1_VCC is required.
USIM1_DET	76	DI	USIM1 detection	The level of this interface is 1.8 V. If hot-swapping function is not used, pull the pin up to the VDD_1P8 with a 47 kΩ resistor.

N706 provides two USIM card interfaces and supports dual-SIM single standby. USIM0 is used first by default. The two interfaces are self-adaptive to 1.8 V/3 V USIM cards. Figure 5-24 shows the reference design of the USIC card interface.

Figure 5-24 Reference design of the USIM card (with hot-swap) interface



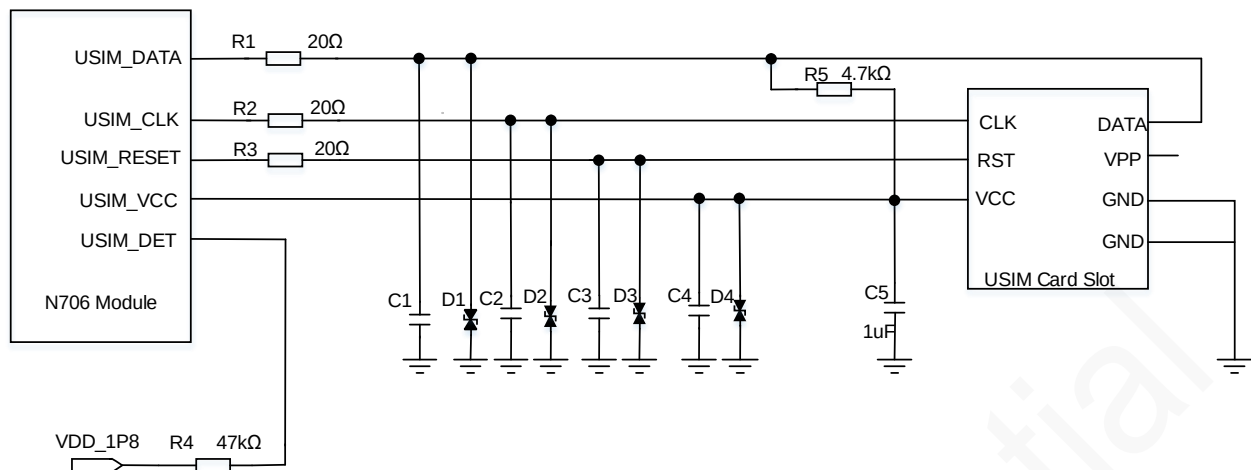
Schematic Design Guidelines

- USIM_VCC is the pin to supply power for USIM card and its maximum load is 50 mA. It is only used as power supply for USIM card (forbidden to supply power to other loads). A filter capacitor must be added on the pin in design.
- Add a pull-up resistor externally to pull up the USIM_DATA pin to USIM_VCC since there is no internal pull-up inside USIM_DATA.
- USIM_CLK is the clock signal pin of the USIM card. In applications with complex electromagnetic environments that have high requirements for ESD protection, it is recommended to add ESD protection diodes (junction capacitance ≤ 7 pF) on each signal cable or add an integrated TVS array.
- Connect a resistor (not greater than 20 Ω) respectively to USIM_DATA, USIM_RESET, USIM_CLK, and USIM_DET (close to the card connector) in series to enhance the ESD performance.
- C1 - C5 are designed to place the high-frequency filter capacitors. The recommended capacitance value is not greater than 10 pF. No capacitors are placed on them by default in design. Please adjust it according to the actual debugging results.
- N706 supports USIM card detection. USIM_DET is a 1.8 V interrupt pin. The USIM detection circuit works by checking the levels across the USIM_DET pin before and after a USIM card is inserted. In the reference circuit, SIM_DET is not connected before a USIM card is inserted and is connected to a 1.8 V voltage after a USIM card is inserted. The high level means USIM card detected while the low level means no USIM card detected.



If the USIM card hot-swapping function is not used, the USIM_DET pin must be pulled up to 1.8 V with a 47 k Ω resistor in series, and the USIM hot-swapping detection function must be disabled in software. Figure 5-25 shows the reference design of USIM card (without hot-swapping function) interface.

Figure 5-25 Reference design of the USIM card (without hot-swap) interface



PCB Design Guidelines

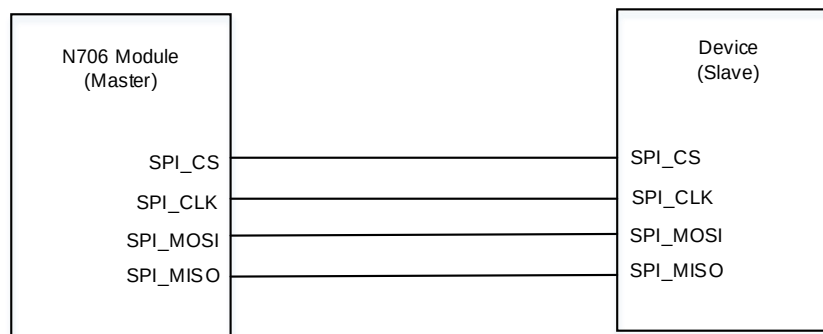
- USIM signals are like to be jammed by RF radiation, resulting in failure to detect the SIM card. Place the USIM card far away from the module
- Place the USIM card close to the module and keep USIM traces as short as possible.
- On the USIM signal, connect the series resistor and ESD protection component close to the USIM card.
- Surround USIM traces with ground to enhance EMC.

5.3.4 SPI

Signal	Pin	I/O	Function description	Remarks
SPI2_MISO	49	DI	SPI output of the slave device and input of the master device	-
SPI2_MOSI	50	DO	SPI input of the slave device and output of the master device	-
SPI2_CS	51	DO	SPI chip select signal of the slave device	-
SPI2_CLK	52	DO	SPI clock	-

The SPI interface supports 1.8 V and only master mode. It supports frequencies up to 50 MHz. The following figure shows the recommended design of the SPI interface.

Figure 5-26 Recommended design of the SPI interface

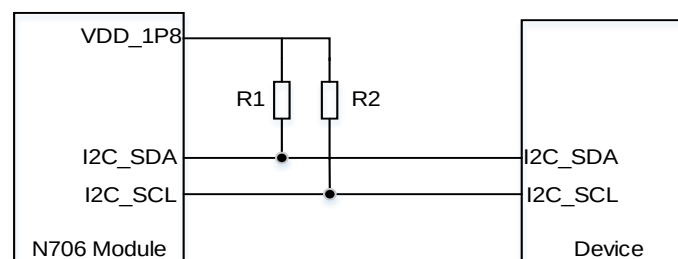


5.3.5 I2C

Signal	Pin	I/O	Function description	Remarks
I2C1_SDA	61	B	I2C data	Connect this pin to VDD_1P8 through a pull-up resistor.
I2C1_SCL	62	DO	I2C clock	Connect this pin to VDD_1P8 through a pull-up resistor.
I2C2_SDA	63	B	I2C data	Connect this pin to VDD_1P8 through a pull-up resistor.
I2C2_SCL	64	DO	I2C clock	Connect this pin to VDD_1P8 through a pull-up resistor.

N706 provides two I2C interfaces with 1.8 V level, supporting only the master mode. Their transmission rates in standard mode and fast mode are respectively 100 Kbps and 400 Kbps. For the pull-up resistor, select a proper resistance based on the actual debugging result; it is recommended to select the resistance ranging from 1.8 k Ω to 10 k Ω . Figure 5-27 shows the reference design of the I2C interface.

Figure 5-27 Reference design of the I2C interface



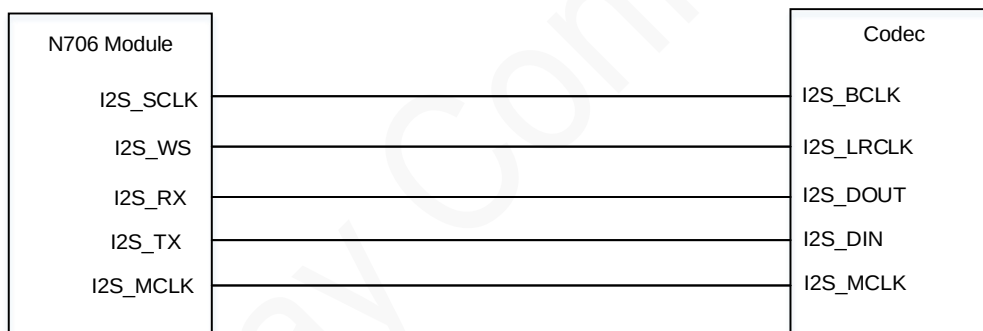
5.4 Audio Interface

N706 provides a group of 1.8 V digit I2S interfaces, which can be multiplexed as PCM interfaces.

Signal	Pin	I/O	Function description	Remarks
I2S_SCLK	30	DO	I2S data clock	Can be multiplexed as PCM_SCLK.
I2S_WS	31	DO	I2S right and left channels selection	Can be multiplexed as PCM_SYNC.
I2S_RX	32	DI	I2S data receiving	Can be multiplexed as PCM_DIN.
I2S_TX	33	DO	I2S data sending	Can be multiplexed as PCM_DOUT.
I2S_MCLK	65	DO	I2S main clock signal	-

The following figure shows the recommended design of the I2S interface.

Figure 5-28 Reference design of the I2S interface



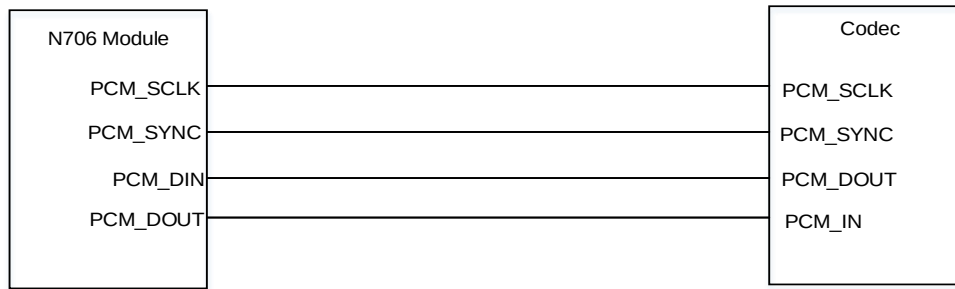
Schematic Design Guidelines

If the levels of N706 and Codec do not match, add the level shifting circuit.

PCB Design Guidelines

- Reduce the cross routing between the I2S signal cable and other cables. If cross routing cannot be avoided, keep this signal cable perpendicular to other cables to reduce coupling.
- Keep the I2S signal cable away from areas where static electricity may be introduced.
- I2S_MCLK must be wrapped by the ground wire. It is also recommended that other signal cables be wrapped with ground wires.

Figure 5-29 PCM reference design



The schematic design and PCB design guidelines of the PCM interface are the same as those of the I2S interface.

5.5 Video Interfaces

5.5.1 LCD

N706 provides an LCD-dedicated SPI interface, supporting frame rates up to QVGA@30fps. The following shows the LCD interface pinouts.

Table 5-2 LCD interface pinouts

Signal	Pin	I/O	Function description	Remarks
LCD_SPI_MOSI	45	DO	SPI input of the slave device and output of the master device	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_CS.
LCD_SPI_MISO	46	DI	SPI output of the slave device and input of the master device	LCD dedicated SPI interface Can be multiplexed as FLASH_SPI_CLK.
LCD_SPI_CS	47	DO	SPI chip select signal of the slave device	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_MISO.
LCD_SPI_CLK	48	DO	SPI clock	LCD-dedicated SPI interface Can be multiplexed as FLASH_SPI_MOSI.
LCD_TE	8	DO	LCD frame synchronization	LCD-dedicated SPI interface Can be multiplexed as FLASH_HOLD.
LCD_RST_N	55	DO	LCD reset	-

5.5.2 Camera

N706 provides a camera-dedicated SPI interface, supporting frame rates up to 0.3M@24fps. The following shows the camera interface pinouts.

Table 5-3 Camera interface pinouts

Signal	Pin	I/O	Function description	Remarks
AVDD_CAM	58	PO	Camera power supply	The power supply is disabled by default. After it is enabled, it outputs 1.8 V/100 mA by default.
CAM_PWDN	66	DO	Power down control	-
VDD_CAMIO	81	PO	CAM IO power supply	The power supply is disabled by default. After it is enabled, it outputs 1.8 V/100 mA by default.
CAM_SPI_MISO	83	B	SPI output of the slave device and input of the master device	Camera-dedicated SPI interface
CAM_SPI_MOSI	84	B	SPI input of the slave device and output of the master device	
CAM_SPI_CLK	85	DO	SPI clock	
CAM_RST_N	86	DO	SPI reset	
CAM_MCLK	87	DO	CAM main clock	-

Schematic Design Guidelines

Notably, AVDD_CAM and VDD_CAMIO are the power supply pins for cameras, with a maximum load capacity of 100 mA. Do Not use these two pins to supply power for other loads. A filter capacitor must be added on the pin in design.

5.6 Other Interfaces

5.6.1 ADC

The module provides two 12-bit ADC channels, and the input voltage ranges from 0.1 V to VBAT. They can be used to perform temperature detection or other related detections. For details, see the *Neoway_N706_AT_Commands_Manual*.

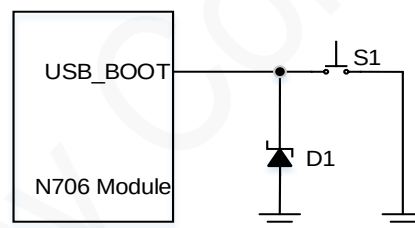
Signal	Pin	I/O	Function description	Remarks
ADC1	9	AI	Universal analog-to-digital signal	-
ADC2	56	AI	Universal analog-to-digital signal	-

5.6.2 USB_BOOT

Signal	Pin	I/O	Function description	Remarks
USB_BOOT	53	DI	Forcible download pin	-

USB_BOOT is the pin used to forcibly enter the download mode. To enter forcible download mode you should connect USB_BOOT of the module to ground and then power up the module. This is the last method to handle issues that result in startup or running failures. It is recommended to reserve this pin (as a button or a test point) to facilitate software upgrades and debugging. The following figure shows the reference design of this pin. Adding an ESD component to protect USB_BOOT in the circuit is required.

Figure 5-30 Reference design of USB force download



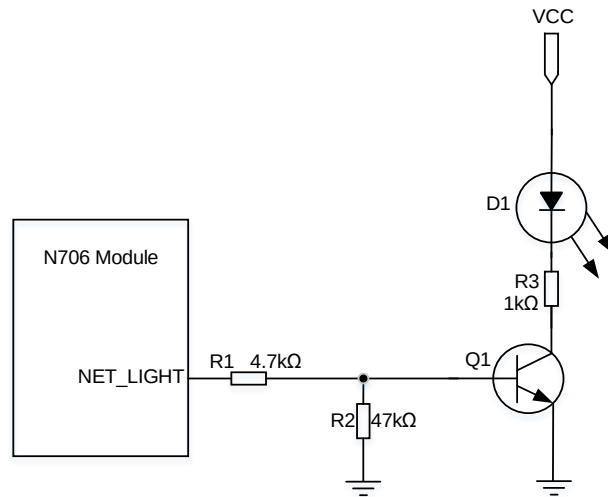
5.6.3 NET_LIGHT

Signal	Pin	I/O	Function description	Remarks
NET_LIGHT	16	DO	Network status indication	Leave this pin floating if it is not used.

NET_LIGHT is the network status indicator pin of the module. It outputs PWM waves of duty cycle varying with the status of the module and drives an LED indicator to blink at different frequencies. You can use the AT command to enable the LED indicator to blink in different states. For detailed usage, see the *Neoway_N706_AT_Commands_Manual*.

Do Not use RING to drive the LED indicator directly since it outputs a high level of 1.8 V. It is recommended to drive the LED indicator by controlling a triode.

Figure 5-31 Driving LED indicator with a triode



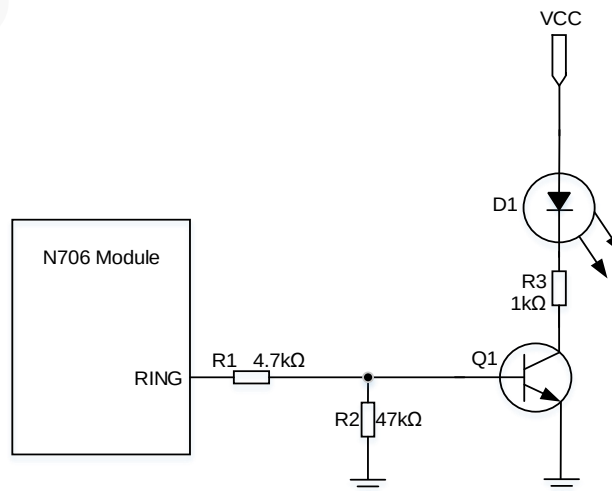
5.6.4 RING

Signal	Pin	I/O	Function description	Remarks
RING	25	DO	Incoming call indicator control	Leave this pin floating if it is not used.

RING is the incoming call indicator control pin of the module. It outputs PMW waves of duty cycle varying with the status of the module and drives an LED indicator to blink at different frequencies. You can use the AT command to enable the LED indicator to blink in different states. For detailed usage, see the *Neoway_N706_AT_Commands_Manual*.

Do Not use RING to drive the LED indicator directly since it outputs a high level of 1.8 V. It is recommended to drive the LED indicator by controlling a triode.

Figure 5-32 Driving LED indicator with a triode



5.6.5 GPIO



All GPIO interfaces support multiplexing. You can multiplex the GPIO pin as the other supported function according to the specific application scenario. Contact Neoway FAEs if necessary.

Signal	Pin	I/O	Function description	Remarks
GPIO1	20	B	GPIO with interrupt	Leave this pin floating if it is not used.
GPIO2	21	B	GPIO with interrupt	Leave this pin floating if it is not used.
GPIO3	69	B	GPIO with interrupt	Leave this pin floating if it is not used.
GPIO0	82	B	GPIO with interrupt	Leave this pin floating if it is not used.

N706 provides 4 GPIO interfaces, all of which have the interrupt function.

5.7 RF Interfaces

Signal	Pin	I/O	Function description	Remarks
ANT_GNSS	4	AI	GNSS antenna	
ANT_WLAN_2G	6	AI	2.4 GHz WLAN antenna	50 Ω impedance control
ANT_MAIN	35	AIO	Main antenna	

5.7.1 ANT_MAIN Antenna Interfaces

The antenna interface of the N706 requires the 50 Ω impedance feature. The impedance of the cable from the module interface to the antenna needs to be kept at 50 Ω impedance to ensure RF performance. A matching network needs to be added in the middle. The matching network is generally divided into three types: L type, T type, and π type. As shown in the figure below, the π -type network is recommended.

Figure 5-33 L-type network

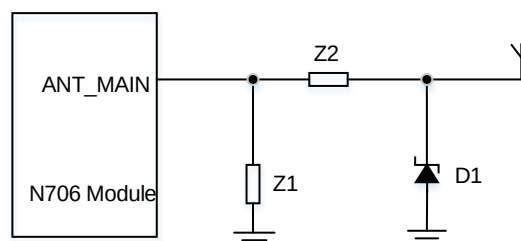
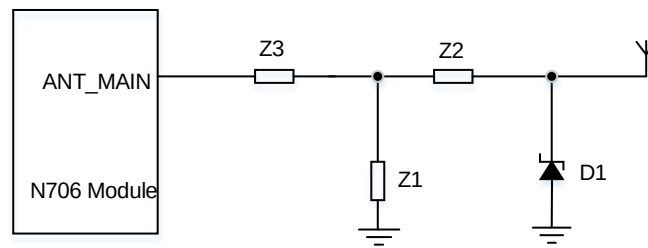
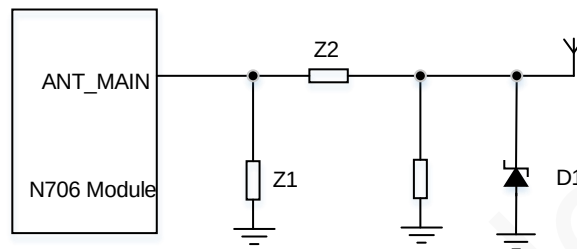


Figure 5-34 T-type network

Figure 5-35 π -type network

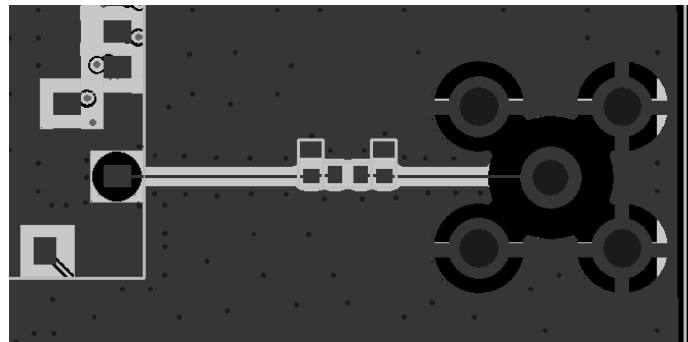
Schematic Design Guidelines

- Element components in the above figures are capacitors, inductors, and $0\ \Omega$ resistors. Place these RLC components as close to the antenna interface as possible.
- If static electricity is introduced at the antenna, it is recommended that electrostatic protection be added. You can use an ESD component with ultra-low junction capacitance. It is recommended that you use an ESD component with a junction capacitance less than $0.5\ \text{pF}$. In addition, you must ensure that the reverse breakdown voltage of the ESD component is greater than $10\ \text{V}$. ESD components with a reverse breakdown voltage of $15\ \text{V}$ or more are recommended.

PCB Design Guidelines

- Lay copper foil around the RF connector. Dig as many ground holes as possible on the copper to ensure $50\ \Omega$ impedance in the trace.
- The trace between the RF pin and the antenna connector should be as short as possible. Control the trace impedance to $50\ \Omega$.
- If you adopt an SMA connector, a big RF solder pad might result in great parasitic capacitance, which will affect the antenna performance. Remove the copper on the first and fourth layers or all layers of a multiple-layer PCB under the RF solder pad.

Figure 5-36 Recommended RF PCB design



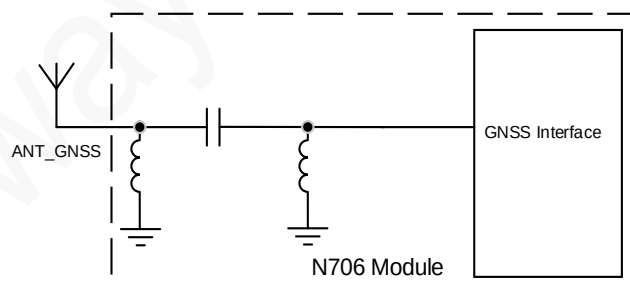
- A reasonable distance should be kept among ANT_MAIN, ANT_WLAN_2G and ANT_GNSS to avoid mutual interference that may affect reception performance.
- On the PCB, keep the RF signals and components far away from high-speed circuits, power supplies, transformers, great inductors, the clock signal, etc.

5.7.2 ANT_GNSS Antenna Interface

GPS Impedance Control

ANT_GNSS is the GNSS RF interface of the module, which requires a characteristic impedance of 50 Ω . Figure 5-37 shows the GNSS structure inside the module.

Figure 5-37 GNSS RF structure



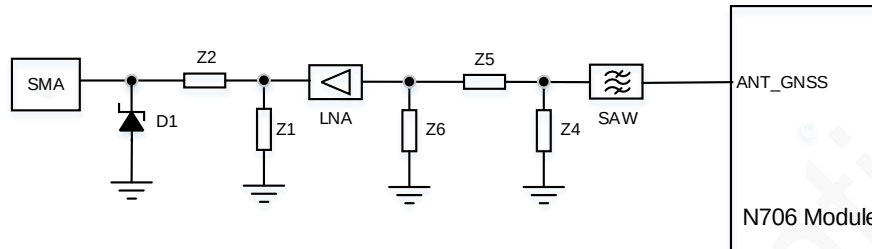
Schematic Design Guidelines

- For the impedance matching circuit between the N706 module and the GNSS antenna, refer to section 5.7.1 “ANT_MAIN Antenna Interfaces”.
- Place an LNA near the feeder because the module does not embed one internally. The module's GPIO interfaces can be used to enable external LNA.

Reference design of passive GNSS antenna

After the GNSS antenna receives the GNSS satellite signal, it is transmitted to the ANT_GNSS pin of the N706 module through the PCB trace. The following figure shows a reference design of the passive GNSS antenna.

Figure 5-38 Reference design of passive GNSS antenna



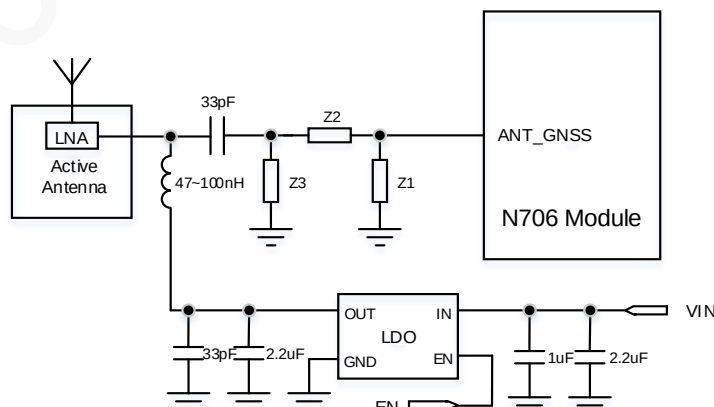
PCB Design Guidelines

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in 5.7.1 “ANT_MAIN Antenna Interfaces”.
- Keep the GNSS antenna circuit far away from the main and WLAN antenna circuits on PCB. Otherwise, these two parts will jam each other, lowering the RF performance.

Reference design of active GNSS antenna

After the antenna receives GNSS satellite signals, the LNA amplifies the signals first and then transmits them to the ANT_GNSS of N706 through feeder and PCB traces. The following figure shows a reference design of the active GNSS antenna.

Figure 5-39 Reference design of the active GNSS antenna



PCB Design Guidelines

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in 5.7.1 “ANT_MAIN Antenna Interfaces”.
- 50 Ω impedance is required for the feeder and PCB traces. Ensure the trace not be too long. The power supply of the active antenna is fed by the 47 nH to 100 nH inductance through the signal traces.
- The output voltage of the LDO used to supply power to the active antenna is related to the selection of the LNA. You can select the appropriate LDO to supply power to the antenna according to the selected LNA device. As shown in Figure 5-39.
- Keep the GNSS antenna circuit far away from the main and WLAN antenna circuits on PCB. Otherwise, these two parts will jam each other, lowering the RF performance.

5.7.3 ANT_WLAN_2G Antenna Interface

The WLAN radio interface only supports the 2.4 GHz frequency band, and its impedance characteristic is required to be 50 Ω .

For the schematic design and the PCB design of the WLAN antenna interface, see the requirements in section 5.7.1 “ANT_MAIN Antenna Interfaces”.

5.7.4 Antenna Assembling

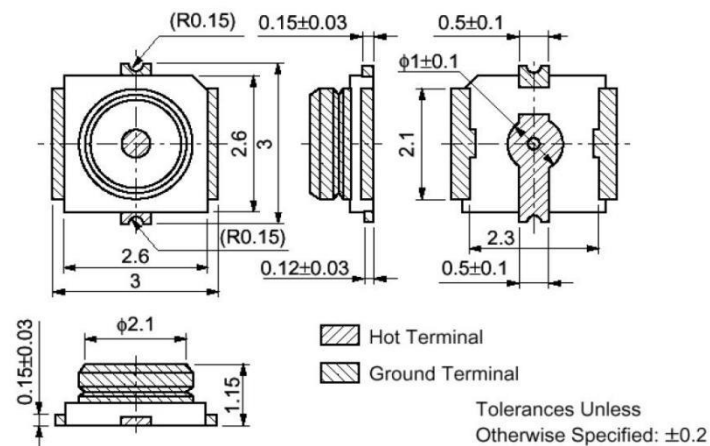
The antenna used by the module must comply with the mobile device standard. The standing wave ratio should be between 1.1 and 1.5, and the input impedance should be 50 Ω . Requirements for antenna gain vary according to the application environment. You can choose a suitable antenna according to specific application scenarios and environments.

Antenna interfaces can be connected to a rubber ducky antenna, magnet antenna, or embedded Planar Inverted F Antenna (PIFA). Keep external RF wires far away from all disturbing sources, especially digital signals and DC/DC power if using RF wires.

The following methods are commonly used to assemble antenna:

- GSC RF connector
MM9329-2700RA1 from Murata is recommended. The following figure shows its encapsulation specifications.

Figure 5-40 Murata RF connector encapsulation specifications



- The RF cable is connected to the module by means of soldering. However, this method has the stability, consistency, and RF performance degradation issues, and therefore is not recommended.

The following figure shows the effect of the connection methods.

Figure 5-41 RF cable connections

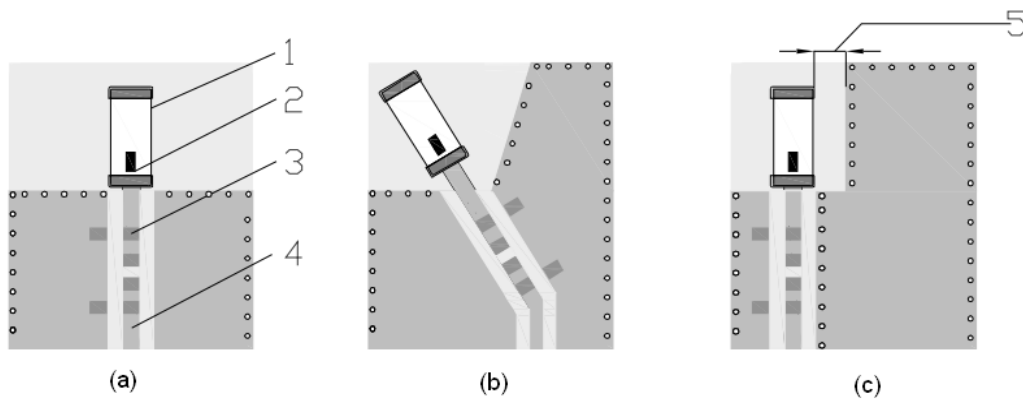


- PCB Printing or SMT

The module works in a wide frequency range, but it is difficult for PCB antennas or ceramic antennas to cover a wide frequency. Therefore, this connection method is recommended only for 2.4 GHz Wi-Fi or BT/BLE antennas.

The following figure shows the layout of the 2.4 GHz ceramic chip antenna. SLDA52-2R540G-S1TF is used as an example.

Figure 5-42 Antenna layout

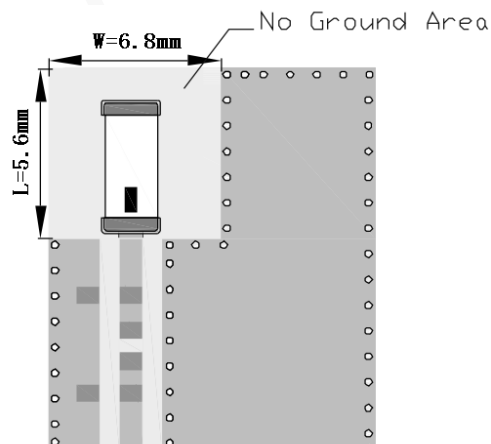


If your PCB is large enough, you can adopt the layout shown in Figure 5-42 (a).

- 1 Chip antenna
- 2 Feeding mark
- 3 Layout pad of the matching circuit
- 4 50 Ω transmission line

Figure 5-43 shows the layout for the area between the antenna and ground that is marked as "5" in Figure 5-42.

Figure 5-43 Layout around the antenna



For more details, refer to the antenna manual and instruction documentation.

6 Electrical Characteristics and Reliability

This chapter describes the electrical characteristics and reliability of the N706 module, including the input and output voltage and current of the power supply, current consumption of the module in different states, operating and storage temperature range, and ESD protection characteristics.

6.1 Electrical Characteristics



If the voltage is lower than threshold, the module might fail to start. If the voltage is higher than threshold or there is a voltage burst during the startup, the module might be damaged permanently.

Table 6-1 N706 operating conditions

Parameter		Minimum value	Typical value	Maximum value
VBAT	V _{in}	3.4 V	3.6 V	4.2 V
	I _{in}	N/A	N/A	TBD

Table 6-2 Current consumption of N706 (typical)

Frequency Band \ Status	Sleep (mA)	Idle (mA)	Active (mA)
FDD-LTE: B1, B3, B5, B8	TBD	TBD	TBD
TDD-LTE: B34, B39, B38, B40, B41	TBD	TBD	TBD

6.2 Temperature Characteristics

Parameter	Minimum value	Typical value	Maximum value
Operating	-30°C	25°C	75°C
Extended*	-40°C*	25°C	85°C*
Storage	-40°C	25°C	90°C



If the module works in an environment where the temperature exceeds the thresholds of the operating temperature range, some of its RF performance indicators might be worse but it can still work properly.

6.3 ESD Protection

Electronic products generally need to undergo strict ESD testing. The following is the ESD protection capability of the main pins of the module. When designing related products, customers need to add corresponding ESD protection according to the industry where the product is used to ensure product quality.

Test environment: humidity 45%; temperature 25°C

Table 6-3 ESD protection

Testing point	Contact discharge	Air discharge
GND	TBD	TBD
ANT	TBD	TBD
Cover	TBD	TBD
Other interfaces	TBD	TBD

7 RF Characteristics

N706 supports the FDD-LTE (Cat.1) and TDD-LTE (Cat.1) network modes. This chapter describes the RF characteristics of N706.

7.1 Operating Band

Table 7-1 N706 operating bands

Operating band	Uplink	Downlink
FDD-LTE B1	1920 - 1980 MHz	2110 - 2170 MHz
FDD-LTE B3	1710 - 1785 MHz	1805 - 1880 MHz
FDD-LTE B5	824 - 849 MHz	869 - 894 MHz
FDD-LTE B8	880 - 915 MHz	925 - 960 MHz
TDD-LTE B34	2010 - 2025 MHz	2010 - 2025 MHz
TDD-LTE B38	2570 - 2620 MHz	2570 - 2620 MHz
TDD-LTE B39	1880 - 1920 MHz	1880 - 1920 MHz
TDD-LTE B40	2300 - 2400 MHz	2300 - 2400 MHz
TDD-LTE B41	2535 - 2655 MHz	2535 - 2655 MHz

7.2 TX Power and RX Sensitivity

Table 7-2 N706 RF transmit power

Band	Max Power	Min. Power
FDD-LTE B1	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B3	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B5	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B8	23 dBm $\pm$ 2 dB	<-40 dBm
TDD-LTE B34	23 dBm $\pm$ 2 dB	<-40 dBm
TDD-LTE B38	23 dBm $\pm$ 2 dB	<-40 dBm

TDD-LTE B39	23 dBm±2 dB	<-40 dBm
TDD-LTE B40	23 dBm±2 dB	<-40 dBm
TDD-LTE B41	23 dBm±2 dB	<-40 dBm

Table 7-3 RF RX sensitivity of N706

Band	Receiving sensitivity
FDD-LTE B1	≤-96.3 dBm
FDD-LTE B3	≤-95 dBm
FDD-LTE B5	≤-95 dBm
FDD-LTE B8	≤-95 dBm
TDD-LTE B34	<-99 dBm
TDD-LTE B38	≤-98 dBm
TDD-LTE B39	≤-99 dBm
TDD-LTE B40	≤-98 dBm
TDD-LTE B41	≤-98 dBm



The preceding indicators are tested in a shielded environment in the laboratory. The LTE band indicators are the test results when the bandwidth is 10 MHz, the modulation mode is QPST and RB is set according to the protocol. On no-shielded environments, deviations may exist in the receiver sensitivity of some individual bands due to the interference.

7.3 N706 GNSS Technical Parameters

Table 7-4 GNSS technical parameters

Parameter	Notice
GPS L1 operating frequency	1575.42±1.023 MHz
GLONASS operating frequency	1597.5 - 1605.9 MHz
BDS operation frequency	1559.1 - 1563.1 MHz
Tracking sensitivity	TBD
Acquisition sensitivity	TBD
Positioning precision (in air)	TBD
Hot start (in air)	TBD

Cold start (in air)	TBD
Update frequency	TBD
Max. positioning altitude	TBD
Max. positioning speed	TBD
Max. positioning acceleration	TBD
CNRin/CNRout	TBD
GNSS data type	TBD
GNSS data type	Passive/active antenna



The tracking sensitivity, and recapture sensitivity are obtained from the signaling test on GSS7000. The values are the maximum values obtained from multiple measurements performed on samples. External LNA was used during the test.

7.4 WLAN Technical Parameters

Table 7-5 WLAN TX power and RX sensitivity

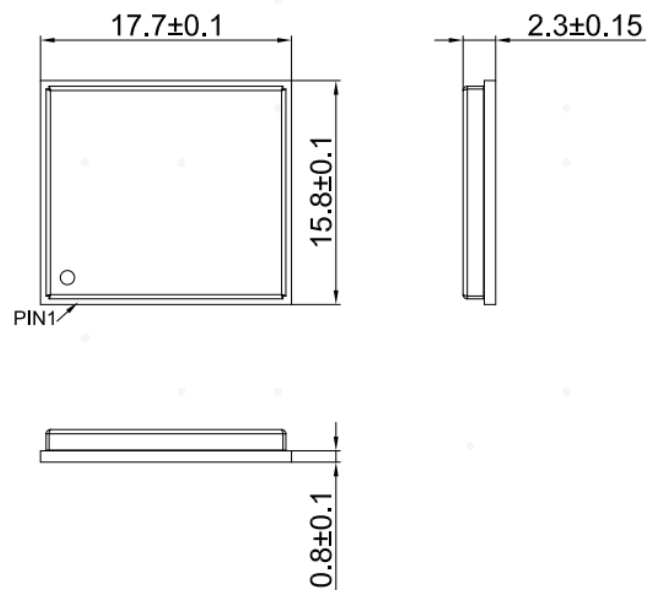
Operating band	Rate	Transmit power	Receiving sensitivity
802.11b (2.4G)	1/2/5.5/11 Mbps	TBD	TBD

8 Mechanical Characteristics

This chapter describes mechanical characteristics of the N706 module.

8.1 Dimensions

Figure 8-1 Top and side view dimensions (unit: mm)



8.2 Label

The label is laser etched, and can withstand a high temperature of 260°C.

Figure 8-2 N706 label



The picture above is only for reference.

8.3 Packing

N706 modules are packed in sealed bags on delivery to guarantee a long shelf life. Follow the same package of the modules again in case of opened for any reason.

8.3.1 Reels

N706 in mass production is delivered in the following packaging.



8.3.2 Moisture

N706 is a level 3 moisture-sensitive electronic element, in compliance with IPC/JEDEC J-STD-020 standard.

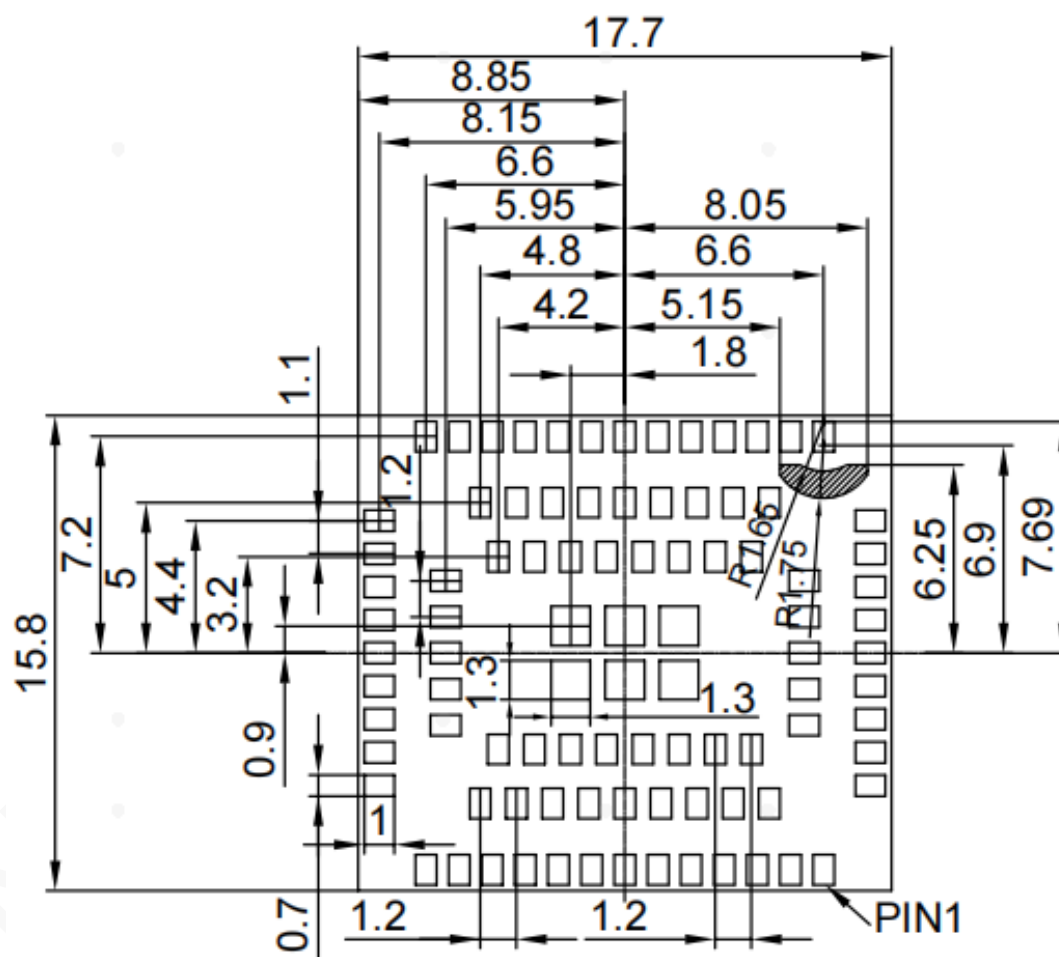
After the module is unpacked, if it is exposed to the air for a long time, the module will get damped, and may be damaged during reflow soldering or laboratory soldering. Bake it before mounting the module. The baking conditions depend on the moisture degree. It is recommended to bake the module at a temperature higher than 90 degrees for more than 12 hours. Do not bake modules with the package tray directly.

9 Mounting

This chapter describes the package of N706, the recommended footprint of the application PCB, and SMT specifications.

9.1 N706 Foot Print

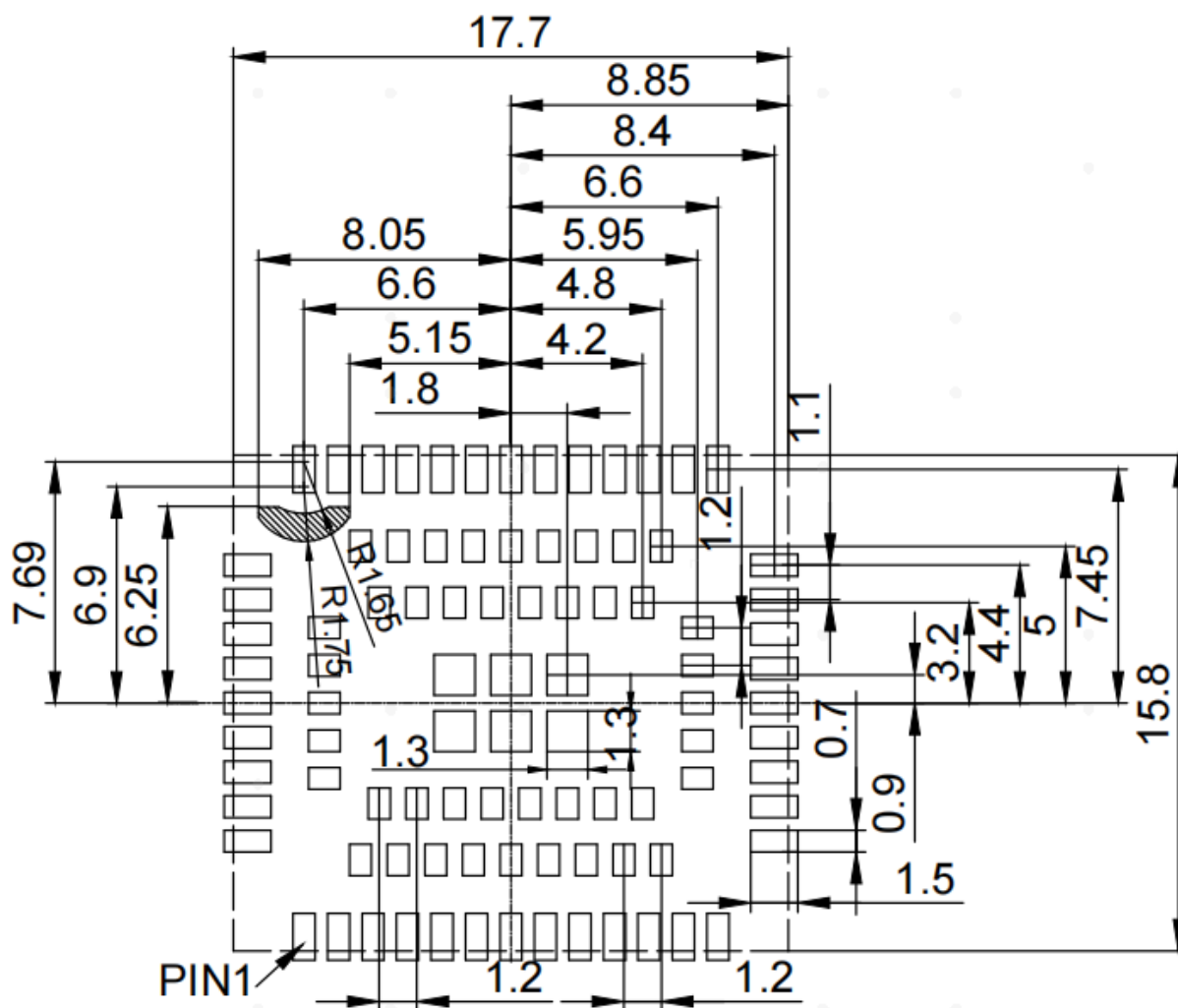
Figure 9-1 N706 foot print (bottom view, unit: mm)



9.2 Application Foot Print

The N706 module has a total of 94 pins in LGA package, and the recommended application PCB package is as follows:

Figure 9-2 N706 PCB foot print (top view, unit: mm)



9.3 Stencil

The recommended stencil thickness is at least 0.15 mm to 0.20 mm.

9.4 Solder Paste

The thickness of the solder paste and the flatness of the PCB are essential for the production yield.

It is recommended to use the same kind of leaded solder paste used during the production process of Neoway.

- The melting point of the leaded solder paste is 35°C lower than that of the lead-free solder paste, and the temperature in the reflow process parameters is also lower than that of the lead-free solder paste. Therefore, the soldering time is shorter accordingly, which easily causes a false solder because LGA in the module is in a semi-melted state during the secondary reflow.
- When using only solder pastes with lead, please ensure that the reflow temperature is kept at 220 °C for more than 45 seconds and the peak temperature reaches 240 °C.

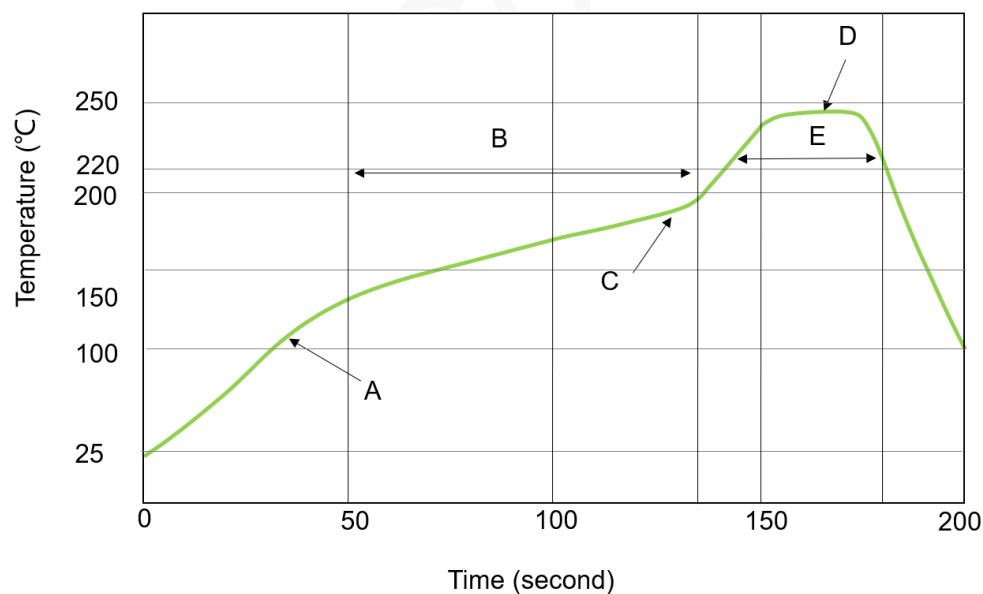
9.5 SMT Oven Temperature Profile



Neoway will not provide warranties for heat-responsive element abnormalities caused by improper temperature control.

Thin or long PCB might bend during SMT. So, use loading tools during the SMT and reflow soldering process to avoid poor solder joint caused by PCB bending.

Figure 9-3 Oven temperature profile



Technical parameters:

- Ramp up rate: 1 to 4 °C/sec
- Ramp down rate: -3 to -1 °C/sec
- Soaking zone: 150-180 °C, Time: 60-100s

- Reflow zone: >220 °C, Time: 40-90s
- Peak temperature: 235-245°C

For information about cautions in storage and mounting, refer to *Neoway_Reflow_Soldering_Guidelines_For_Surface-Mounted_Modules*.

When manually desoldering the module, use heat guns with great opening, adjust the temperature to 245 °C (depending on the type of the solder paste), and heat the module till the solder paste is melted. Then remove the module using tweezers. Do not shake the module at high temperatures while removing it. Otherwise, the components inside the module might get misplaced.

A Abbreviations

Abbreviation	Full name
ADC	Analog-to-Digital Converter
AI	Analog Input
AO	Analog Output
AIO	Analog Input/Output
ARM	Advanced RISC Machine
BT	Bluetooth
bps	Bits per Second
CCC	China Compulsory Certification
CEP	Circular Error Probable
CNR	Carrier to Noise Rate
CS	Chip Select
CTS	Clear to Send
DC	Direct Current
DCS	Digital Cellular System
DI	Digital Input
DIO	Digital Input/Output
DL	Downlink
DO	Digital Output
DPSK	Differential Phase Shift Keying
DQPSK	Differential Quadrature Phase Shift Keying
DRX	Discontinuous Reception
DTR	Data Terminal Ready
ECM	Ethernet Control Model
eDRX	Extended DRX
EGSM	Enhanced GSM
ESD	Electronic Static Discharge
ESR	Equivalent Series Resistance

EVK	Evaluation Kit
FDD	Frequency Division Duplexing
FPC	Flexible Printed Circuit
FTP	File Transfer Protocol
GFSK	Gauss Frequency Shift Keying
GLONASS	GLOBAL NAVIGATION SATELLITE SYSTEM
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input Output
3GPP	3rd Generation Partnership Project
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit
IO	Input/Output
ISP	Image Signal Processor
LCC	Leadless Chip Carriers
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
MCLK	Main Clock
MCU	Microcontroller Unit
MIC	Microphone
PCB	Printed Circuit Board
PCS	Personal Communications Service
PWM	Pulse Width Modulation
QVGA	Quarter Video Graphics Array
RAM	Random Access Memory
RF	Radio Frequency
ROM	Read-only Memory
RTC	Real Time Clock
SD	Secure Digital
SDIO	Secure Digital Input Output

SPK	Speaker
SPI	Serial Peripheral Interface
TDD	Time Division Duplex
UART	Universal Asynchronous Receiver-Transmitter
UL	Uplink
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Battery Voltage
VSWR	Voltage Standing Wave Ratio
Wi-Fi	Wireless Fidelity
WCDMA	Wide-band Code Division Multiple Access
WCI	Wireless Coexistence Interface
WLAN	Wireless Local Area Network